Strategic Search in Patenting: An Empirical Investigation of Patent Breadth

Nick Dunagan*

September 29, 2026

Abstract

I examine how the threat of post-grant patent review affects behavior during patent prosecution. The America Invents Act expanded administrative procedures for challenging issued patents, changing the consequences of obtaining patents vulnerable to later review. I construct an extended dataset linking U.S. published patent applications to granted patents through 2020 and exploit differences in pre-reform litigation exposure across USPTO art units in an exposure-based difference-in-differences design. I find that more exposed art units have narrower minimum independent claims following the reform. A one-standard-deviation increase in historical litigation exposure is associated with approximately 2.25 additional words in minimum independent claim length at publication. Similar patterns appear using first independent claim length. Grant-stage outcomes exhibit substantial pre-existing differences across more and less exposed art units, and a matched-sample decomposition shows that these differences historically arose primarily during prosecution, while in later post-reform cohorts more of the exposure relationship is already present at publication. These findings suggest that changes in downstream challenge risk can affect patenting behavior earlier in the prosecution process.

*John E. Walker Department of Economics, Clemson University, SC 29634 (email: nddunag@g.clemson.edu)

1 Introduction

Patent rights are shaped by incentives that extend beyond the examination stage. The breadth of a patent determines how much protection an invention receives and how much technological space remains available to competitors and follow-on innovators [13, 23, 32]. Yet the value of that breadth depends partly on what happens after a patent is granted. Broad claims may be more valuable when they survive, but they may also face greater scrutiny from competitors with incentives to challenge the patent. Changes to the institutions governing post-grant challenges can therefore affect the returns to the claims applicants seek before examination ever begins.

The America Invents Act provides a natural setting for studying this connection. The AIA created new administrative procedures for challenging issued patent claims, most importantly inter partes review before the Patent Trial and Appeal Board. These procedures changed the environment in which issued patents could be contested and made post-grant review an important part of the U.S. patent system [19, 20]. Existing work shows that post-grant scrutiny can affect behavior earlier in the patenting process. Martinelli and Mazzei find that applicants filed narrower U.S. claims following the AIA and that examiners imposed additional narrowing when applicants did not fully adjust themselves [22]. More generally, models of endogenous patent challenge imply that downstream review can affect incentives even for patents that are never challenged [30]. These results raise a further question. Did the response to the AIA depend on how important patent challenges had historically been in the technological environment where an application was examined?

This paper studies whether applicants facing greater historical exposure to patent litigation changed the breadth they initially sought following the AIA. I measure pre-AIA litigation exposure at the four-digit examiner-art-unit level using utility patents granted from 2003 through 2011 and whether those patents were involved in district-court patent litigation during the same historical period. Examiner art units provide relatively fine technological and examination environments within the U.S. Patent and Trademark Office. The empirical design compares changes in patent outcomes after the AIA across art units with different levels of predetermined litigation exposure. The resulting variation is continuous rather than a division between treated and untreated technologies. The central empirical question is whether applications in historically more litigation-exposed art units changed differently after the common institutional reform.

I measure patent breadth directly from the language of independent claims. Claim language is

particularly useful because patent claims define the legal boundaries of the right, and additional limiting language generally corresponds to narrower protection [14, 21]. The primary outcome is the word length of the shortest independent claim, which I observe in published applications and in eventual patent grants. I also examine the first independent claim and the number of independent claims. Observing claims at both publication and grant is important because these stages capture different parts of the patenting process. Published claims provide a measure of the breadth entering prosecution, while granted claims reflect the combination of initial drafting and subsequent examination. Existing evidence shows that prosecution can substantially narrow patent claims, making these stages economically distinct [21].

To construct these outcomes, I extend claims-based patent measures through 2020 using USPTO full-text records for published applications and patent grants. The resulting data contain more than six million published applications and more than seven million patent grants, with approximately 3.9 million matched publication-grant pairs. I link these records to USPTO examination data and district-court patent litigation records. The preferred analysis restricts attention to four-digit examiner art units containing at least 500 patent grants during the pre-AIA exposure period, leaving 427 art units. Historical exposure is adjusted for differences in the grant-year composition of art units and standardized so that the empirical estimates correspond to a one-standard-deviation increase in pre-AIA litigation exposure.

The main specification is a continuous-exposure difference-in-differences design. I estimate patent outcomes at the examiner-art-unit by filing-cohort level with examiner-art-unit and filing-cohort fixed effects. The interaction between historical litigation exposure and the post-AIA period measures whether outcomes changed more after the reform in art units with greater historical exposure to litigation. Filing cohorts from 2003 through 2011 form the pre-period, 2012 is treated as a transition cohort, and the post-period begins in 2013. Event-study specifications allow the relationship between historical exposure and patent outcomes to vary by filing cohort and provide evidence on the evolution of the exposure gradient before and after the AIA.

The clearest response appears in the claims applicants initially publish. A one-standard-deviation increase in historical litigation exposure is associated with approximately 2.25 additional words in the shortest independent claim after the AIA. The corresponding estimate for the first independent claim is approximately 2.33 words. Because additional language generally adds limitations to an independent claim, these estimates correspond to narrower initially requested patent breadth in

historically more litigation-exposed art units. Independent claim count provides complementary evidence. The post-AIA event-study estimates are consistently negative, indicating that applications in more exposed art units also tend to contain fewer independent claims after the reform. The claim-length estimates are statistically significant and remain positive across a range of alternative art-unit sample restrictions.

The timing of the response supports an interpretation centered on initial applicant behavior. The event-study coefficients for publication-stage claim length fluctuate during the pre-AIA period rather than displaying a sustained movement toward the later post-AIA pattern. Following the transition period, the minimum-independent-claim estimates become consistently positive and reach roughly 2.6 to 2.9 additional words in several later filing cohorts. The pre-treatment evidence is not uniformly clean across every diagnostic, so I report the full event-study paths rather than relying on a single formal pre-trend test. The distinction between the pre-period fluctuations and the persistent post-AIA coefficients nevertheless suggests that the pooled estimate is not simply generated by a smooth differential trend already underway before the reform.

The same pattern does not appear as clearly in the claims that ultimately receive patent protection. Estimated post-AIA effects on minimum and first independent claim length at grant are small and statistically imprecise, and the grant-stage event studies display stronger pre-existing relationships between historical litigation exposure and claim length. Publication-to-grant changes also exhibit differential pre-period dynamics that complicate a causal interpretation of prosecution-stage effects. These results therefore provide much stronger evidence about the breadth applicants initially seek than about a change in final granted breadth. The contrast is economically informative because it suggests that the response to the post-AIA challenge environment appears early in the patenting process and can be attenuated as applications proceed through examination.

The paper contributes to the literature on patent breadth and post-grant review by showing that exposure to downstream disputes predicts how strongly applicants respond to a common institutional reform. Existing claims-based research establishes that patent breadth can be measured from independent-claim language and that breadth changes substantially during prosecution [14, 21]. Research on examination shows that the rights emerging from prosecution respond to examiner incentives and institutional constraints [5, 4]. Most closely related, Martinelli and Mazzei use U.S. and Canadian patent twins to estimate the average effect of stronger U.S. post-grant review on applicant and examiner behavior [22]. I instead use predetermined differences in historical litigation exposure

within the U.S. patent system to study where the response to the AIA was strongest. The results show that the shift toward narrower initial claims is concentrated in examination environments where patent litigation had previously been more salient.

The results also provide empirical support for a broader view of patenting as a sequential strategic process. Applicants choose claims while anticipating examination and the value of the rights that may eventually be granted. Examiners subsequently evaluate and modify those claims, while potential challengers decide whether issued rights are worth contesting. A change in downstream challenge institutions can therefore alter behavior upstream even when a particular patent is never challenged. The evidence in this paper is consistent with that feedback operating through applicant claim drafting. Historically greater exposure to patent disputes is associated with a larger post-AIA movement toward narrower claims at publication, while the corresponding relationship is substantially less clear by the time a patent reaches grant.

The paper proceeds as follows. Section 2 describes patent claims, prosecution, and the changes to post-grant review introduced by the AIA. Section 3 describes the conceptual framework used in the analysis. Section 4 places the paper within the literature on patent breadth, examination, and post-grant challenge. Section 5 describes the construction of the claims, examination, and litigation data and the historical litigation-exposure measure. Section 6 presents the empirical design. Section 7 reports the main publication, grant, and prosecution results. Section 8 examines sample robustness, exposure support and non-linearity, and matched-stage evidence. Section 9 discusses the interpretation of the results, and the final section concludes.

2 Institutional and Policy Background

Patent rights are produced through a sequential administrative process that begins with an applicant's proposed claims and ends with the claims, if any, that survive examination. A nonprovisional patent application contains a specification describing the invention and one or more claims defining the protection sought. The claims are particularly important because they establish the boundaries of the legal right that the applicant is asking the Patent Office to grant [50, 49]. Independent claims stand on their own, while dependent claims incorporate and further limit another claim. Applicants may initially seek relatively broad protection, but the claims can be amended as the application proceeds through examination. Amendments must remain supported by the original disclosure because

applicants generally cannot add new matter after filing.

Most U.S. patent applications become public before examination is complete. Subject to statutory exceptions, the USPTO publishes an application approximately eighteen months after the earliest filing date for which priority is claimed [39]. The publication contains the specification and claims available to the Office for publication and therefore provides an observable measure of the protection sought relatively early in the patenting process. Published claims should not be interpreted as a perfect copy of the claims on the original filing date. Applicants may submit preliminary amendments before the first Office action, and those amendments can be reflected in the patent application publication when they are received early enough to be incorporated [47, 40]. Publication nevertheless occurs well before the final outcome for many applications and provides a useful distinction between the claims presented early in prosecution and the claims that eventually issue.

Patent prosecution begins after filing and allows both the applicant and the examiner to affect the claims that may ultimately be granted. Applicants can submit prior-art references through information disclosure statements, respond to examiner objections with legal arguments and amendments, provide supporting evidence, and participate in examiner interviews that become part of the prosecution record [36, 46]. Applicants and their representatives owe a duty of candor and good faith requiring disclosure of known information that is material to patentability. This obligation does not create a general requirement that applicants conduct a patentability search. The USPTO expressly states that filing an information disclosure statement does not represent that a search has been conducted and that applicants are not generally required to perform such a search [41, 43]. The information initially available to the examiner can therefore depend partly on decisions made by applicants before and during prosecution.

Examiners independently evaluate the application and search for relevant prior art. After determining the breadth of the claims, an examiner searches the prior art and considers both references identified independently and references submitted by the applicant [44, 48]. The examiner evaluates the claims under the statutory requirements for patentability, including novelty and nonobviousness, and communicates objections and rejections through Office actions. Applicants can respond by presenting arguments, supplying evidence, or amending the claims. The examiner then reconsiders the application in light of the updated record [45]. This iterative process can continue through multiple rounds until claims are allowed, prosecution ends in abandonment, or the applicant pursues further administrative procedures.

The resulting granted patent can differ substantially from the application observed at publication. Claims may be amended by adding limitations, claims may be cancelled, and some applications never reach grant. The empirical stages used in this paper therefore correspond to economically different points in the patenting process. Published claims capture the breadth sought relatively early in prosecution. The difference between publication and grant reflects changes occurring while the application is prosecuted, although those changes can result from applicant decisions, examiner actions, or their interaction. Granted claims describe the final breadth that survives this process. This distinction is central to the empirical analysis because a change observed in published claims has a different institutional interpretation from a change that appears only after examination.

Issued patents remain subject to challenge after leaving the Patent Office. Patent validity can be contested in federal court when patent rights are asserted, and administrative mechanisms for reconsidering issued patents existed before the America Invents Act. Ex parte reexamination allowed the USPTO to reconsider patentability in light of qualifying prior art, while inter partes reexamination provided a mechanism for third parties to participate in the reconsideration of certain issued patents. These procedures meant that grant did not fully resolve uncertainty about the enforceable breadth of a patent. District-court litigation remained an especially important setting in which competitors and other parties could contest the validity and breadth of issued rights.

The Leahy-Smith America Invents Act substantially changed this post-grant environment. Signed into law on September 16, 2011, the AIA was a broad reform of the U.S. patent system that changed both pre-grant and post-grant institutions [34]. Among its post-grant provisions, the Act replaced inter partes reexamination with inter partes review and created post-grant review, with the new proceedings conducted before the Patent Trial and Appeal Board [42, 37]. The new administrative proceedings became available on September 16, 2012. These changes created a more prominent adversarial mechanism through which third parties could ask the Patent Office to reconsider the patentability of claims after issuance.

Inter partes review provides the most broadly relevant new post-grant procedure for the patents studied here. A party other than the patent owner can petition the PTAB to review one or more issued claims on novelty or obviousness grounds based on prior art consisting of patents or printed publications [38]. The Board may institute review when the petition demonstrates a reasonable likelihood that the petitioner would prevail with respect to at least one challenged claim. IPR took effect on September 16, 2012 and can apply to patents issued before, on, or after that date. Its

reach therefore extended immediately to a large stock of existing patent rights rather than only to applications filed after enactment.

Post-grant review created a related but broader procedure. A PGR petition can raise a wider range of patentability grounds and must generally be filed within nine months after a patent issues [52]. Unlike IPR, however, PGR generally applies to patents arising from applications governed by the AIA's first-inventor-to-file provisions. Those provisions became effective on March 16, 2013 [35]. PGR therefore became relevant more gradually as patents originating under the new filing regime began to issue. For much of the period immediately surrounding implementation, IPR represented the more immediate change in administrative post-grant exposure for the existing stock of U.S. patents.

The AIA changed additional features of the patent system at roughly the same time. The Act moved the United States from a first-to-invent system toward a first-inventor-to-file regime, changed aspects of the prior-art rules, authorized third-party preissuance submissions, and introduced supplemental examination, among other provisions [34, 51, 35]. These provisions did not all become effective on the same date. Several examination and post-grant provisions, including IPR and PGR, took effect on September 16, 2012, while the first-inventor-to-file provisions took effect on March 16, 2013. The implementation period therefore represented a broader institutional transition rather than a single policy change occurring on one filing date.

This timing motivates treating applications filed during 2012 separately in the empirical analysis. Applications in that cohort were filed while important provisions of the AIA were being implemented, and the new post-grant procedures became available only during the final months of the year. Beginning with later filing cohorts provides a clearer distinction between applications entering the patenting process before and after the new challenge regime was in operation. The empirical analysis consequently uses filing cohorts through 2011 as the pre-period, treats 2012 as a transition cohort, and begins the post-period in 2013.

The institutional changes created by the AIA need not have had the same economic importance across all areas of patenting. Technologies differed substantially in how frequently patents had historically become involved in downstream disputes, and applicants operating in those environments entered the reform with different experiences of patent litigation and challenge. The next section uses this heterogeneity to motivate historical litigation exposure as a measure of how salient the changed post-grant environment was likely to be across examination settings.

3 Conceptual Framework

The empirical analysis is motivated by the sequential strategic-search framework developed in the companion paper, *Strategic Search in Patenting* [3]. The model treats patent breadth as the outcome of interactions among an applicant, an examiner, and a potential challenger. The applicant acquires information and chooses the breadth carried into prosecution, the examiner adds information and determines how much of that breadth survives examination, and a potential challenger observes the resulting patent and prosecution record before deciding whether additional search and a post-grant challenge are worthwhile. The central implication for this paper is that these stages are connected. A change in the incentives facing a downstream challenger can affect applicant and examiner behavior before a patent is ever granted.

The America Invents Act provides a natural institutional change through which to examine this mechanism. The introduction of inter partes review created an administrative alternative for challenging issued patent claims and therefore changed the expected environment facing patent owners after grant. The closest analogue in the companion model is a shift in the distribution of challenger filing costs, G_C, toward a lower-cost environment. This interpretation is deliberately reduced form. The AIA changed several parts of the patent system at approximately the same time, as discussed in Section 2, and the empirical analysis does not claim to identify a change in a single structural parameter. The useful conceptual implication is that the reform increased the importance of potential downstream challenge when applicants and examiners made earlier decisions.

Stronger downstream challenge incentives first affect the applicant's problem. In the model, lowering the effective cost of challenge increases the value of resolving uncertainty before a patent is issued. The applicant responds by acquiring more information and pursuing breadth that is more likely to remain defensible if challenged. The empirical analogue is a movement toward narrower claims early in prosecution. Because additional limiting language generally corresponds to narrower independent claims, this mechanism predicts longer minimum and first independent claims in the published application [14, 21]. A reduction in the number of independent claims can provide complementary evidence when fewer independent claiming routes correspond to a more constrained set of rights.

The applicant response can also change what occurs during examination. Examiner search in the companion model depends on the record inherited from the applicant. When the applicant acquires

more information before or early in prosecution, the examiner receives a stronger information state and the marginal return to additional examiner search falls. Under the partial-adjustment condition in the theory paper, examiner search s_E can therefore decline when applicant search s_A rises without completely undoing the applicant's initial response [3]. This interaction implies that stronger applicant adjustment before examination can reduce the amount of additional narrowing required later in prosecution.

The distinction between publication and grant is therefore important for interpreting the empirical results. Published claims provide the closest observable measure of the breadth carried into prosecution before the full examination process has occurred. Changes between publication and grant reflect subsequent prosecution, including responses by the applicant, the examiner, or both. Granted claims combine the initial applicant decision with these later adjustments. If applicants respond to stronger downstream challenge incentives by seeking narrower claims initially, the effect may be most visible at publication. Examination can then partially offset that initial difference because applications entering prosecution with more constrained claims require less additional adjustment. The model therefore does not require the publication-stage and grant-stage responses to be equal in magnitude.

This stagewise interpretation is especially useful for the empirical design because the three outcomes answer different questions. A post-AIA increase in independent claim length at publication is consistent with applicants entering prosecution with narrower claims. A smaller publication-to-grant increase in claim length is consistent with less additional narrowing being required during examination. A narrower granted claim reflects the cumulative effect of both stages. The strongest theoretical prediction concerns the applicant-side response. The precise magnitude remaining at grant depends on how strongly examiner behavior adjusts to the applicant's earlier choices and on how claim-language measures map into the model's underlying breadth object.

Other downstream outcomes have less definite predictions. Lower filing costs make challenges directly more attractive to potential challengers. At the same time, applicant and examiner responses can produce narrower and better-supported patents before grant, reducing the value of challenging the rights that eventually issue. These forces operate in opposite directions, so the companion model does not generally sign the equilibrium change in observed challenge activity. The absence of an increase in litigation or administrative challenges would therefore not rule out the upstream mechanism studied here. The cleaner empirical implications concern how applicants and examiners

adjust before the downstream challenge decision is reached.

The empirical design uses historical litigation exposure to capture where these incentives should have mattered most. The AIA changed federal patent institutions broadly, but its economic importance need not have been uniform across technologies. In some examination environments, issued patents had historically been much more likely to become involved in district-court disputes. Applicants operating in these areas therefore had greater prior exposure to the possibility that the rights they obtained would later be contested. A reform that made post-grant challenge more accessible should have greater consequences for expected patent value in these environments than in areas where downstream disputes had historically been rare.

Historical litigation exposure is therefore a measure of the preexisting salience of downstream challenge rather than the treatment itself. The treatment is the common institutional change generated by the AIA. Exposure measures how strongly the incentives created by that change should vary across examiner art units. The empirical analysis consequently asks whether post-AIA changes in applicant and prosecution behavior are systematically larger in examination environments with greater pre-AIA litigation exposure. The detailed construction of this measure is described in Section 5.

This interpretation also clarifies the relationship between the empirical paper and *Strategic Search in Patenting*. The empirical analysis does not estimate the structural parameters of the model or attempt to test every comparative static. Instead, it evaluates a central implication of the model using an institutional change and observable patent outcomes. If downstream challenge incentives feed backward into earlier patenting decisions, a change in the post-grant environment should alter the claims applicants seek before examination, particularly where downstream challenge had historically been more salient. Publication-stage claim breadth provides the clearest test of that mechanism, while prosecution changes, granted breadth, and citation behavior provide evidence on how the response develops across later stages of the patenting process.

4 Related Literature

Patent breadth is an economically important margin of patent protection. Early theoretical work emphasizes that the strength of patent protection depends on the breadth of the exclusionary right as well as its duration [6, 13]. Broader protection can increase the returns to invention, but it can

also restrict competition and reduce the opportunities available to subsequent innovators. These concerns are particularly important when innovation is cumulative and later inventions build directly on earlier discoveries [23, 32, 27]. The resulting tradeoff makes patent breadth an important policy margin rather than a secondary feature of an issued patent. Empirical work similarly finds that differences in patent breadth are associated with economically meaningful outcomes, including the private value of patented inventions [18]. This literature provides the basic motivation for studying whether patent breadth responds to changes in the institutions governing patent rights.

Measuring patent breadth has become increasingly tied to the language that legally defines the patent right. Early empirical work relied on technological classifications, citation measures, claim counts, and related patent characteristics to proxy for economically important differences across patents [18, 15, 26]. These measures are useful for studying patent value and technological importance, but they do not necessarily correspond closely to the legal boundaries established by the claims. More recent work therefore develops measures directly from claim language. Kuhn and Thompson use the length of the first claim as a measure of breadth and validate its interpretation using assessments by patent attorneys and applicant behavior [14]. Marco, Sarnoff, and deGrazia similarly use independent-claim length and independent-claim count to characterize the breadth of patent protection [21]. Additional limiting language generally narrows the range of subject matter covered by an independent claim, making longer independent claims a useful measure of narrower patent breadth when applications are compared within sufficiently similar technological environments. These measures also allow patent breadth to be observed when an application is published and again when a patent is granted, which makes it possible to study how breadth changes during prosecution.

The distinction between application and grant breadth matters because patent rights are jointly shaped by applicants and the Patent Office. Applicants choose the claims with which they enter prosecution, while examiners evaluate those claims against the prior art and determine whether they can be allowed in their existing form. Empirical work documents substantial heterogeneity in examiner behavior and shows that institutional constraints can influence this process. Examiner characteristics predict grant outcomes and subsequent patent characteristics [16, 29]. Frakes and Wasserman show that examination time affects examiner scrutiny and granting behavior, with reductions in allotted examination time increasing the likelihood that applications are granted [5]. Feng and Jaravel show that examiner-driven variation can have consequences after grant, includ-

ing differences in litigation, invalidation, acquisition by patent assertion entities, and subsequent innovation [4]. Marco, Sarnoff, and deGrazia provide complementary evidence directly from claim language, showing that independent claims generally narrow between publication and issuance [21]. Together, these results imply that the breadth observed at grant reflects both the claims initially sought by applicants and the subsequent prosecution process.

Applicants also adjust their patenting strategies in response to institutional incentives. The decision problem extends beyond whether to seek a patent because applicants can change how they draft claims, acquire information, disclose prior art, and continue prosecution. Hegde, Mowery, and Graham document the strategic use of continuing applications within the U.S. patent system [12], while Righi, Cannito, and Vladasel show that continuations are used to modify the breadth of patent protection and respond to information revealed during prosecution [28]. Changes in patent institutions can also shift applicant behavior along margins other than breadth. Bekkers, Martinelli, and Tamagni study an expansion of prior-art information at the European Patent Office and find lower grant rates without a corresponding reduction in the breadth of the patents that are ultimately granted [2]. Hegde, Herkenhoff, and Zhu show that earlier publication following the American Inventors Protection Act altered the timing and direction of follow-on innovation [11]. These studies show that institutional changes can alter patenting behavior well before a final patent right is observed, while leaving open which particular margin adjusts in response to a given reform.

Post-grant challenges provide one channel through which downstream institutions can affect behavior earlier in the patenting process. Patent validity is uncertain, and the Patent Office cannot identify every relevant piece of prior art or fully resolve every question of validity during initial examination [17]. This concern motivated proposals for administrative post-grant review well before the America Invents Act. Hall, Graham, Harhoff, and Mowery argue that a post-grant opposition system could improve the information available for evaluating issued patents and provide a less costly means of reconsidering questionable rights [9]. Graham and Harhoff use evidence from European opposition proceedings to evaluate the potential benefits of such a system in the United States [8]. More generally, Schankerman and Schuett show that examination and ex post challenges jointly determine the screening of patent quality, while the selection of patents into challenges limits the extent to which downstream review can correct weaknesses in initial examination [30]. This literature establishes a direct link between the quality of initial examination and the incentives created by later opportunities to challenge issued rights.

Evidence from existing post-grant systems suggests that the prospect of later review can also discipline behavior before a challenge occurs. European opposition provides an important setting for studying this mechanism because patents have long been subject to centralized post-grant challenges at the European Patent Office. Harhoff and Reitzig show that opposition is selective and disproportionately targets economically important patents [10]. Nagler and Sorg provide evidence of a broader disciplinary response, finding that inventors whose marginal patents are invalidated subsequently reduce low-quality patenting [25]. The creation of inter partes review under the America Invents Act brought a stronger administrative challenge mechanism to the United States, and subsequent work documents substantial selection into these proceedings [20, 1]. The economic consequences of post-grant review can therefore extend beyond the set of patents that ultimately appear before a review board. A credible threat of later scrutiny can alter expected patent value while an application is still being drafted and examined.

Martinelli and Mazzei provide the closest empirical evidence that the AIA changed patent breadth through this disciplinary channel [22]. They compare U.S. and Canadian applications covering the same underlying inventions and use the introduction of stronger U.S. post-grant review to study responses by applicants and examiners. Their results indicate that U.S. applicants entered prosecution with narrower claims following the reform and that examination generated additional narrowing when applicants did not fully adjust themselves. They also find that these responses are weaker in technologies characterized by dense patent thickets, where incentives surrounding the breadth of any individual patent may differ. Their analysis provides evidence that changes in downstream scrutiny can alter patent breadth before a patent is ever challenged and establishes the closest empirical precedent for the mechanism studied here.

This paper studies the same broad connection between downstream challenge incentives and upstream patenting behavior using a different source of variation. I measure the historical exposure of U.S. examiner art units to district-court patent litigation before the AIA and ask whether patenting behavior changed differentially after the reform in examination environments where downstream disputes had previously been more common. Historical litigation exposure captures differences in the preexisting salience of patent challenges across technologies while the AIA provides a common change in the institutional environment. The resulting design complements the international comparison in Martinelli and Mazzei by using variation within the U.S. patent system to identify where the response to post-grant reform was strongest.

The application-stage evidence further distinguishes the paper from work that observes only the rights that survive examination. The largest and most consistent changes appear in the independent claims contained in published applications, where historically more litigation-exposed art units experience larger post-AIA increases in claim length. The corresponding estimates for granted claims are substantially weaker. This distinction links the literature on claims-based measurement to work showing that applicants and examiners respond endogenously to patent institutions [21, 5, 4, 22]. The results are most consistent with a response in the breadth applicants initially seek, which can subsequently be altered or attenuated as applications move through examination.

5 Data and Measurement

5.1 Patent Data and Claims Measures

I construct a patent-level dataset combining published U.S. patent applications, issued patents, and USPTO examination records. Claim text comes from the USPTO's weekly full-text files for patent application publications and patent grants. Published application data begin in 2001, following the introduction of routine pre-grant publication in the United States, and I reconstruct application- and grant-stage claim measures through 2020. I supplement these records with the USPTO Patent Examination Research Dataset (PatEx), which provides application identifiers, filing and disposition dates, examiner information, examiner art-unit assignments, and other prosecution characteristics [7, 24]. The resulting data extend the claims-based measures developed in the USPTO Patent Claims Research Dataset beyond its original 2014 endpoint [21].

The reconstructed application data contain 6,318,649 unique pre-grant publications from 2001 through 2020, of which 6,318,594 link to PatEx using application identifiers. The grant data contain 7,092,807 patents issued between 1976 and 2020. Linking publications and grants through application identifiers produces 3,903,452 matched publication–grant records. Combining publication-only applications, grant-only patents, and matched publication–grant records yields a complete lifecycle master containing 9,508,004 observations.

I reconstruct claim-level information directly from the USPTO full-text files. For each publication and grant, the parser separates the document into individual claims and records the claim number, claim text, and independent or dependent status. For published applications, it also identifies canceled claims. Introductory claim language such as "I claim" and "What is claimed is" is removed,

and the displayed claim numeral is retained as an identifier but excluded from the word count. Claim dependency is identified using structured claim references supplemented by textual references where necessary. Canceled application claims are retained during parsing but excluded from the document-level breadth measures.

I construct three principal claims-based measures. The primary outcome is *minimum independent claim length*, defined as the word count of the shortest non-canceled independent claim in the document. The second is *first independent claim length*, defined as the word count of the first non-canceled independent claim appearing in the document. The third is *independent claim count*, defined as the number of non-canceled independent claims.

Independent claims define enforceable rights without incorporating limitations from another claim. Adding words to an independent claim generally adds elements or limitations that must be satisfied for infringement, making the claim narrower in breadth. Longer independent claims are therefore interpreted as narrower claims, while shorter independent claims are interpreted as broader claims [14, 21]. Minimum independent claim length focuses on the shortest, and generally broadest, independent claim in the document and is my preferred measure of the outer boundary of claimed protection. First independent claim length provides a closely related alternative, while independent claim count captures a different dimension of claiming strategy. I therefore treat independent claim count as a complementary measure rather than a strictly monotonic measure of patent breadth [21].

The claim measures are observed separately at publication and grant. For applications observed at both stages, I also construct publication-to-grant changes. Claim-length changes equal grant-stage length minus publication-stage length, so a positive value indicates that the relevant independent claim becomes longer, and therefore generally narrower, during prosecution. Independent-claim-count changes are defined analogously.

5.2 Claims Validation

Because the reconstructed claim measures extend beyond the period covered by the historical USPTO claims data, I validate the parser using 2014, the final year of overlap [21]. The validation compares reconstructed and historical records at both the claim and document levels and is conducted separately for published applications and patent grants.

The reconstruction closely reproduces the historical data. For published applications, 6,725,663 historical claims are matched, corresponding to more than 99.999 percent coverage of the comparable

historical claims universe. Independent-versus-dependent classification agrees in 99.99989 percent of matched claims and word counts are within one word in 99.99635 percent. For grants, 4,978,864 reconstructed utility-patent claims match the historical data, corresponding to 99.6767 percent coverage of reconstructed utility-patent claims available for comparison. Independent-versus-dependent classification agrees in 99.99998 percent of matched claims and word counts are within one word in 99.99827 percent.

Because first independent claim length is used directly as an outcome, I also validate it separately. The reconstruction identifies the same first independent claim in 99.998 percent of comparable published applications and 99.999 percent of comparable grants. The resulting first independent claim length is within one word of the historical measure in approximately 99.99 percent of comparisons at both stages. Table 1 reports the headline validation results. Appendix Tables 6 and 7 report the full claim- and document-level validation exercises.

Table 1: Patent Data Construction and Claims Validation

	Count	Percent
Panel A: Patent data construction		
Published applications, 2001–2020	6,318,649	
Published applications linked to PatEx	6,318,594	99.9991
Patent grants, 1976–2020	7,092,807	
Matched publication–grant records	3,903,452	
Complete lifecycle master	9,508,004	
Panel B: 2014 claims validation		
Application claims matched	6,725,663	99.9993
Application independent/dependent classification		99.99989
Application word count within one word		99.99635
Grant claims matched	4,978,864	99.6767
Grant independent/dependent classification		99.99998
Grant word count within one word		99.99827

Notes: Panel A reports counts from the reconstructed patent lifecycle universe. The PatEx match percentage is calculated relative to reconstructed published applications. Panel B compares reconstructed claim records with the historical USPTO claims data for 2014. The grant claim match percentage is a coverage statistic for reconstructed utility-patent claims available for comparison. Classification and word-count agreement are conditional on matched claims. Full validation results are reported in the appendix.

5.3 Historical Litigation Exposure

I measure historical patent disputes using the USPTO Patent Litigation Docket Reports Data, which identify patent cases filed in U.S. federal district courts and the patents associated with those cases [33, 31]. After removing duplicate patent–case combinations and normalizing patent identifiers, I

define a patent as historically litigated if it appears in at least one district-court patent case during 2003–2011. Using an indicator rather than the number of lawsuits prevents a small number of repeatedly litigated patents from disproportionately determining an art unit's exposure.

The litigation records link closely to the ordinary utility-patent identifiers relevant to the exposure construction. Among 26,926 numeric patent identifiers involved in district-court litigation during 2003–2011, 26,716 link to the USPTO patent universe, a match rate of 99.22 percent. Most remaining unmatched identifiers correspond to non-utility patent formats or substantially earlier patent-number ranges. Additional linkage diagnostics are reported in the appendix.

I aggregate historical litigation to four-digit examiner art units. Art units provide relatively narrow technological and examination environments while retaining substantially more cross-sectional variation than broad technology classifications. The measure is intended to capture predetermined differences in the historical importance of downstream patent disputes across examination environments rather than post-AIA litigation outcomes.

Let L_i equal one if patent i was involved in at least one district-court patent case during 2003–2011 and zero otherwise. The raw litigation exposure of examiner art unit a is

$$\text{Exposure}_a^{Raw} = 1000 \frac{\sum_{i \in a} L_i}{N_a},$$

where N_a is the number of eligible utility patents granted in art unit a during 2003–2011. The historical exposure universe contains 1,573,621 four-digit-art-unit grants, of which 10,587 were litigated during the historical period, corresponding to 6.73 litigated patents per 1,000 grants.

A raw litigation rate can partly reflect differences in grant timing because patents granted earlier in the historical window have more time to enter litigation before the window closes. An art unit concentrated in early grant cohorts could therefore appear more litigation-exposed even without a greater underlying litigation propensity. I remove these common grant-cohort differences by estimating

$$L_i = \alpha + \gamma_{y(i)} + \varepsilon_i,$$

where $\gamma_{y(i)}$ denotes grant-year fixed effects. The adjusted art-unit measure is

$$\text{Exposure}_a^{Adj} = 1000 \left(\bar{L} + \bar{\varepsilon}_a \right),$$

where $\bar{L}$ is the overall historical litigation rate and $\bar{\varepsilon}_a$ is the mean residual within art unit a. Adding the overall mean back preserves an interpretable per-1,000-grants scale while removing common grant-year differences in litigation incidence. Because this is a residualized index rather than a literal litigation rate, adjusted exposure can take slightly negative values.

The preferred regression variable standardizes Exposure_a^{Adj} to mean zero and standard deviation one within the preferred examiner-art-unit sample. The unstandardized adjusted measure has a mean of 8.09 and a standard deviation of 6.69 litigated patents per 1,000 historical grants. A one-unit increase in the regression exposure variable therefore corresponds to a one-standard-deviation increase in adjusted historical exposure.

Because district-court litigation is relatively rare, exposure estimates for very small art units can be noisy. I therefore restrict the preferred sample to four-digit art units containing at least 500 eligible historical grants. Of 719 art units observed in the historical universe, 427 satisfy this restriction, retaining 1,543,382 of 1,573,621 historical grants, or 98.08 percent of the four-digit grant population. The baseline does not require an art unit to appear in every individual historical year; alternative size and balance restrictions are considered as robustness checks.

Figure 1 shows the distribution of adjusted historical litigation exposure across the preferred art units. The median is 6.24 litigated patents per 1,000 grants, with a 25th percentile of 3.86 and a 75th percentile of 10.39. The raw and grant-year-adjusted measures are highly correlated at 0.952, indicating that the adjustment does not substantially reorder art units by historical exposure.

Exposure is fixed within examiner art unit and constructed entirely from historical grant and litigation information. It should therefore be interpreted as predetermined cross-sectional heterogeneity in historical litigation exposure rather than as treatment intensity itself. The AIA provides the common institutional change, while exposure determines the dimension along which responses to that change may differ.[1]

5.4 Analysis Sample and Descriptive Statistics

The empirical analysis is organized by application filing cohort. Publication outcomes use filing cohorts from 2003 through 2018, while grant-stage and publication-to-grant outcomes use cohorts from 2003 through 2016 to limit right-censoring in grant data ending in 2020. The 2012 filing cohort is omitted as a transition cohort, with 2003–2011 defining the pre-period and 2013 onward defining

[1]The baseline historical window includes district-court litigation recorded during calendar year 2011.

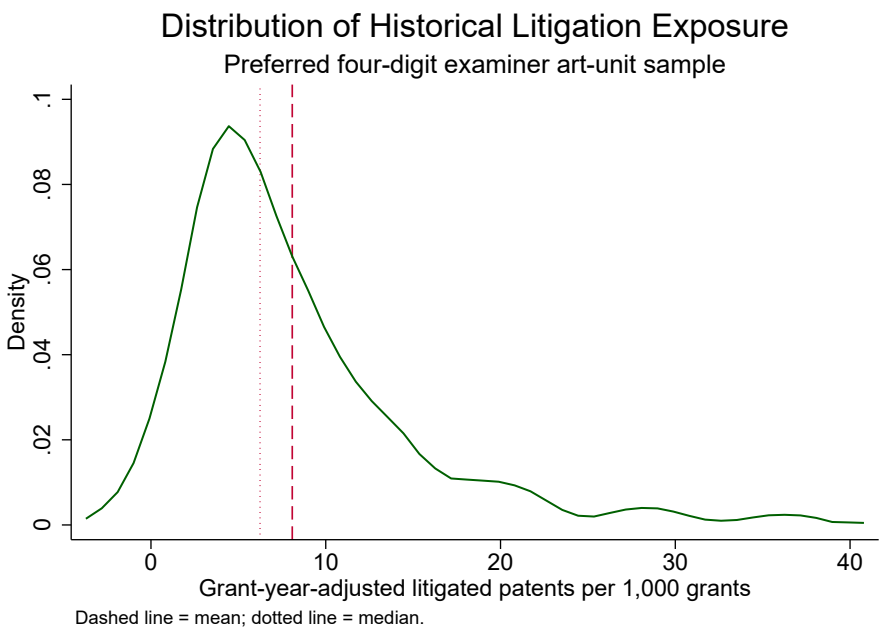


Figure 1: Distribution of Historical Litigation Exposure

Notes: The figure plots grant-year-adjusted historical litigation exposure across the 427 four-digit examiner art units in the preferred sample. Exposure is expressed on a per-1,000-historical-grants scale. Vertical reference lines indicate the mean and median. The regression measure standardizes this index to mean zero and standard deviation one.

the post-period.

The publication sample contains all usable published applications and does not condition on eventual grant. The grant sample contains all usable issued patents regardless of whether a corresponding publication is observed. Publication-to-grant changes are calculated only for applications observed at both stages. I therefore use the publication sample for the headline early-stage outcomes, the grant sample for issued-patent outcomes, and the matched sample only for within-application prosecution changes.

The preferred publication population contains 4,167,068 applications with independent-claim-count information. Minimum independent claim length is observed for 4,164,106 applications and first independent claim length for 4,156,739. The corresponding grant population contains 2,596,109 patents, with minimum and first independent claim length observed for 2,596,100 and 2,596,089 patents, respectively. The matched publication–grant population contains 2,403,286 applications, of which 2,401,596 have observed changes in minimum independent claim length and 2,397,378 have observed changes in first independent claim length.

Table 2 reports descriptive statistics for these samples. At publication, the mean minimum independent claim contains 112.47 words, compared with 167.94 words among issued patents. Among applications observed at both stages, minimum independent claim length increases by an average of 51.91 words and first independent claim length by 49.58 words, while independent claim count declines slightly. Claim-length measures are highly dispersed across patents. These differences are descriptive; comparisons between the unrestricted publication and grant samples should not be interpreted as within-application prosecution changes.

Figure 2 provides a complementary descriptive view of the relationship between historical litigation exposure and minimum independent claim length over filing cohorts. For each filing cohort, I separately regress the relevant claim-length outcome on the standardized historical exposure measure, weighting examiner-art-unit cells by the number of underlying applications or patents. The resulting coefficient is the raw cross-sectional exposure gradient for that filing cohort. Unlike the event-study estimates presented below, these coefficients are not normalized relative to an omitted reference year and do not include examiner-art-unit fixed effects.

The raw gradients reveal substantial differences across stages. The publication-stage exposure gradient is relatively small during much of the earlier sample and becomes more positive in later filing cohorts. By contrast, the grant-stage gradient is already positive in many earlier cohorts.

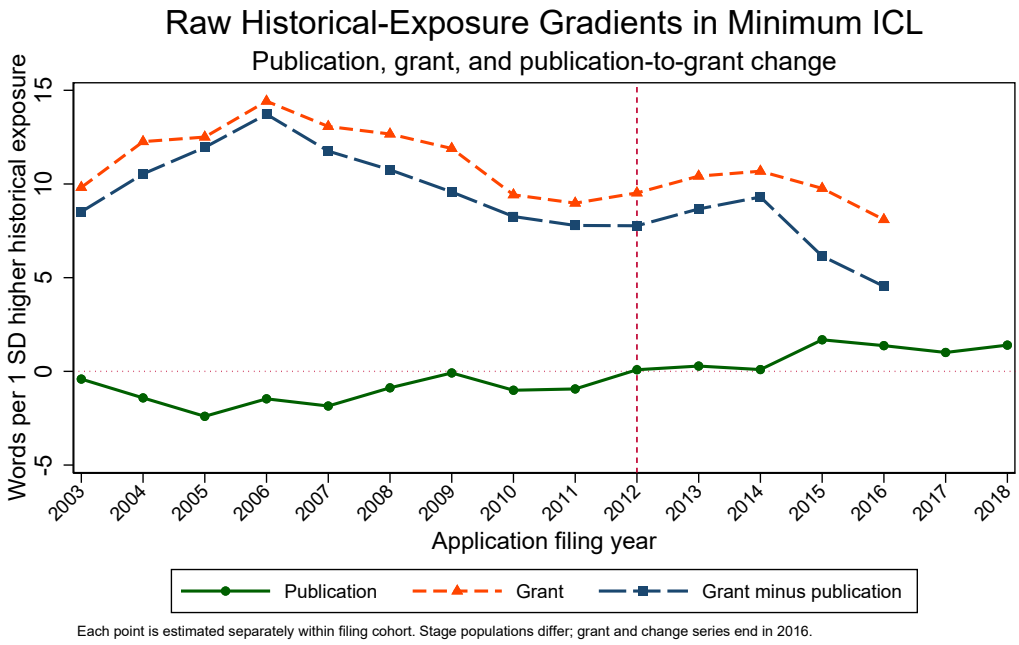


Figure 2: Historical Litigation Exposure and Minimum Independent Claim Length Across Filing Cohorts

Notes: The figure plots the year-specific raw relationship between standardized historical litigation exposure and minimum independent claim length. Each coefficient is estimated separately within an application filing cohort using examiner-art-unit-level outcome means weighted by the number of underlying observations. Publication refers to minimum independent claim length in published applications, Grant refers to minimum independent claim length in issued patents, and Grant minus publication refers to the within-application change among applications observed at both stages. No filing cohort is used as an omitted reference year. The vertical dashed line denotes the 2012 transition cohort. Publication outcomes are observed through 2018, while grant and publication-to-grant outcomes are observed through 2016. The three series are based on different analysis populations and therefore should not be interpreted as an exact decomposition of grant-stage claim length.

Table 2: Descriptive Statistics for Preferred Analysis Samples

	N	Mean	SD	P25	Median	P75
Panel A: Publication sample						
Minimum independent claim length	4,164,106	112.47	113.78	59	92	140
First independent claim length	4,156,739	133.11	135.35	74	110	162
Independent claim count	4,167,068	2.74	2.35	1	2	3
Panel B: Grant sample						
Minimum independent claim length	2,596,100	167.94	111.76	101	147	209
First independent claim length	2,596,089	187.02	120.67	116	165	230
Independent claim count	2,596,109	2.53	1.70	1	2	3
Panel C: Publication-to-grant change						
Change in minimum independent claim length	2,401,596	51.91	102.14	0	32	81
Change in first independent claim length	2,397,378	49.58	107.88	0	30	79
Change in independent claim count	2,403,286	-0.28	2.27	0	0	0

Notes: Statistics are calculated at the individual application or patent level within the preferred 500+ historical-grant examiner-art-unit sample. The publication sample uses application filing cohorts from 2003–2018; the grant and matched change samples use filing cohorts from 2003–2016. The 2012 filing cohort is excluded from all three populations. Claim-length changes equal grant-stage length minus publication-stage length and are calculated only for applications observed at both stages.

The publication-to-grant series shows that a substantial exposure gradient in claim-length changes during prosecution is also present well before the later increase observed at publication. These patterns are descriptive rather than causal, but they help motivate the stage-specific analysis below and underscore why publication, grant, and prosecution outcomes are considered separately.

6 Empirical Strategy

6.1 Continuous-Exposure Difference-in-Differences

The empirical design asks whether patent-claim outcomes changed differentially after the America Invents Act (AIA) in examiner art units with greater historical exposure to downstream patent litigation. The AIA provides a common institutional change, while the historical litigation measure described in Section 5 provides predetermined cross-sectional variation in the extent to which different examination environments had previously been associated with downstream patent disputes.

The baseline analysis is conducted at the four-digit-examiner-art-unit by application-filing-cohort level. For each art unit a and filing cohort t, I calculate the mean outcome among the underlying applications or patents contributing to that cell. I estimate

$$Y_{at} = \alpha_a + \gamma_t + \beta \left(Exposure_a \times Post_t \right) + \varepsilon_{at}, \tag{1}$$

where Y_{at} is the cell-level mean outcome, α_a denotes examiner-art-unit fixed effects, and γ_t denotes application-filing-cohort fixed effects. $Exposure_a$ is the grant-year-adjusted historical litigation exposure measure standardized to have mean zero and standard deviation one across the preferred examiner-art-unit sample. $Post_t$ equals one beginning with the 2013 filing cohort, and the 2012 filing cohort is excluded as a transition cohort.

Regressions are weighted by the number of underlying applications or patents contributing to each art-unit-by-cohort cell. These weights preserve the relative contribution of the underlying observations while allowing the estimation to be conducted on the substantially smaller art-unit-by-cohort panel.

The coefficient β measures how the post-AIA change in the outcome varies with historical litigation exposure. Because exposure is standardized, β is the differential post-AIA change associated with a one-standard-deviation increase in historical exposure. For a claim-length outcome, for example, a positive coefficient indicates that claims became relatively longer, and therefore narrower, after the AIA in historically more exposed art units.

This can be viewed as a continuous version of the usual difference-in-differences comparison. Rather than dividing art units into treated and untreated groups, the specification compares changes across the full distribution of historical exposure. Art units with different exposure levels therefore provide different points along a common exposure gradient. Historical exposure itself is not treatment, however; it is fixed before the post-AIA period and captures the dimension along which responses to the common institutional change are allowed to differ.

6.2 Identification

Identification comes from changes over filing cohorts within examiner art units and from whether those changes vary systematically with predetermined historical litigation exposure. Examiner-art-unit fixed effects absorb persistent differences across examination environments, including permanent differences in technologies, examination practices, and average claim characteristics. Filing-cohort fixed effects absorb aggregate changes in patent outcomes that are common across art units.

The identifying assumption is that, absent the AIA, changes in the outcomes would not have been systematically related to historical litigation exposure. Put differently, more- and less-exposed art units may differ in their average claim characteristics, but the relationship between exposure and changes in those characteristics should not shift systematically around implementation of the AIA

for reasons unrelated to the policy.

This assumption does not require art units with different exposure levels to have the same outcome levels. Persistent differences are absorbed by the art-unit fixed effects. It also does not require each art unit to experience identical year-to-year fluctuations. The relevant restriction is instead that differential changes across art units are not systematically associated with historical litigation exposure in a way that would generate the estimated post-AIA gradient independently of the institutional change.

A causal interpretation also requires that other contemporaneous changes did not affect examination environments differentially in ways that are systematically correlated with historical litigation exposure. Changes in the composition of applications across art units could similarly confound the estimates if they occurred around the AIA and were related to historical exposure. The robustness analyses therefore examine alternative definitions of the art-unit sample and historical exposure, while the event-study specification below shows how the exposure gradient evolves across filing cohorts.

The design differs from a staggered-adoption difference-in-differences setting. The AIA occurs at the same time for all examiner art units, so the estimates do not rely on comparing units treated at different dates. Instead, the identifying variation comes from whether the outcome changes differently after a common institutional change according to a predetermined continuous exposure measure.

The baseline specification imposes a linear relationship between historical exposure and the differential post-AIA change. Under this specification, β summarizes the exposure gradient with a single coefficient. Additional specifications using alternative exposure definitions and groupings provide checks on whether the baseline result is sensitive to that functional-form choice.

6.3 Event-Study Specification

I examine the evolution of the exposure gradient more flexibly using an event-study specification. I replace the single post-AIA interaction in Equation (1) with filing-cohort-specific exposure interactions:

$$Y_{at} = \alpha_a + \gamma_t + \sum_{\tau \neq 2011,\, 2012} \beta_\tau \left(Exposure_a \times \mathbf{1}\{t = \tau\} \right) + \varepsilon_{at}. \tag{2}$$

The 2011 filing cohort is the omitted reference cohort and 2012 is excluded as the transition cohort. Each coefficient β_τ therefore measures how the relationship between historical litigation exposure and the outcome in filing cohort τ differs from the corresponding relationship in 2011.

The pre-AIA coefficients provide a visual diagnostic of how stable the exposure gradient was before the institutional change, while the post-AIA coefficients trace how that gradient evolves after implementation. The event study therefore complements the pooled estimate by showing whether the post-AIA relationship appears immediately, develops gradually, or varies across subsequent filing cohorts.

Both the pooled and event-study specifications are estimated using examiner-art-unit-by-filing-cohort cells and analytic weights equal to the number of underlying observations in each cell. Because exposure is defined at the examiner-art-unit level and each art unit contributes observations across multiple filing cohorts, regression errors may be correlated within art units over time. I therefore report heteroskedasticity-robust standard errors clustered by four-digit examiner art unit. The preferred sample contains 427 examiner-art-unit clusters.

The event-study figures report 95 percent confidence intervals based on the same art-unit-clustered standard errors. I use the same inference procedure throughout the pooled and dynamic specifications so that the reported uncertainty corresponds to the level at which the historical exposure measure varies.

7 Main Results

I begin with the event-study estimates because they show how the relationship between historical litigation exposure and claim outcomes evolves across filing cohorts. I then summarize these dynamics using the pooled continuous-exposure difference-in-differences specification from Equation (1). The strongest evidence appears in claims observed at publication. Grant-stage outcomes display substantially more pre-AIA variation in the exposure gradient, while publication-to-grant changes help show how the relationship evolves during prosecution.

7.1 Event-Study Evidence

Figure 3 reports the event-study estimates for minimum and first independent claim length at publication. The coefficients measure the exposure gradient in each filing cohort relative to the

corresponding gradient in the 2011 reference cohort.

For minimum independent claim length, the pre-AIA estimates fluctuate around the 2011 gradient rather than displaying a sustained directional movement toward the later post-AIA estimates. The coefficients range from approximately -1.2 to 1.0 words across the pre-period and are close to zero in several cohorts. Beginning with the 2013 filing cohort, however, every estimate is positive. The estimated differential is approximately 1.3 words in 2013 and 1.2 words in 2014 before increasing to approximately 2.6–2.9 words in each cohort from 2015 through 2018.

First independent claim length exhibits a very similar pattern. The pre-AIA coefficients move on both sides of the 2011 reference gradient. The post-AIA estimates are positive in every cohort, with relatively small estimates in 2013 and 2014 followed by increases to approximately 2.4–3.4 words from 2015 through 2018. The similarity across the two claim-length measures is notable because minimum independent claim length selects the shortest independent claim in the application, whereas first independent claim length follows a claim whose position is determined independently of its eventual length.

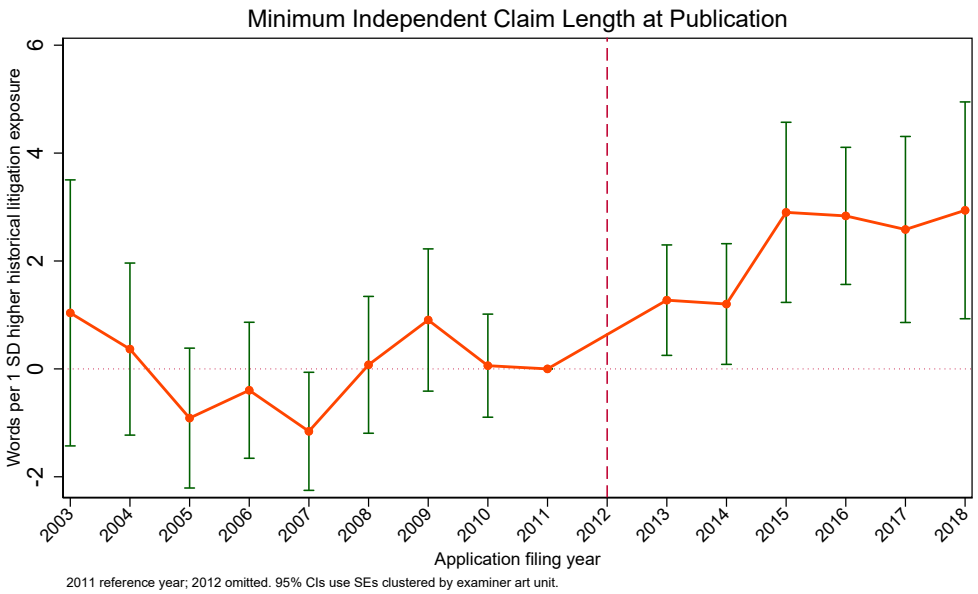
Taken together, the publication event studies show a persistent positive shift in the exposure gradient after the AIA, with the difference becoming particularly pronounced beginning with the 2015 filing cohort. Because longer independent claims generally contain additional limitations, the estimates are consistent with relatively narrower initial claims in historically more litigation-exposed examination environments.

Independent claim count provides complementary evidence. The publication event-study coefficients are negative in every post-AIA filing cohort, ranging from approximately -0.02 to -0.03 independent claims per one-standard-deviation increase in historical exposure. The pre-AIA coefficients are generally small and display substantially less systematic movement. Because independent claim count is not a strictly monotonic measure of patent breadth, I treat this evidence as complementary to the claim-length measures. The corresponding event study is reported in the appendix.

Figure 4 presents the analogous estimates for claims observed at grant. The grant-stage dynamics differ substantially from those at publication. For minimum independent claim length, several early pre-AIA cohorts exhibit exposure gradients approximately three to four words larger than the 2011 gradient. These differences decline through the pre-period and are close to zero by 2010. First independent claim length follows nearly the same pattern.

Positive exposure gradients reappear after the AIA. For minimum independent claim length, the

Panel A: Minimum Independent Claim Length



Panel B: First Independent Claim Length

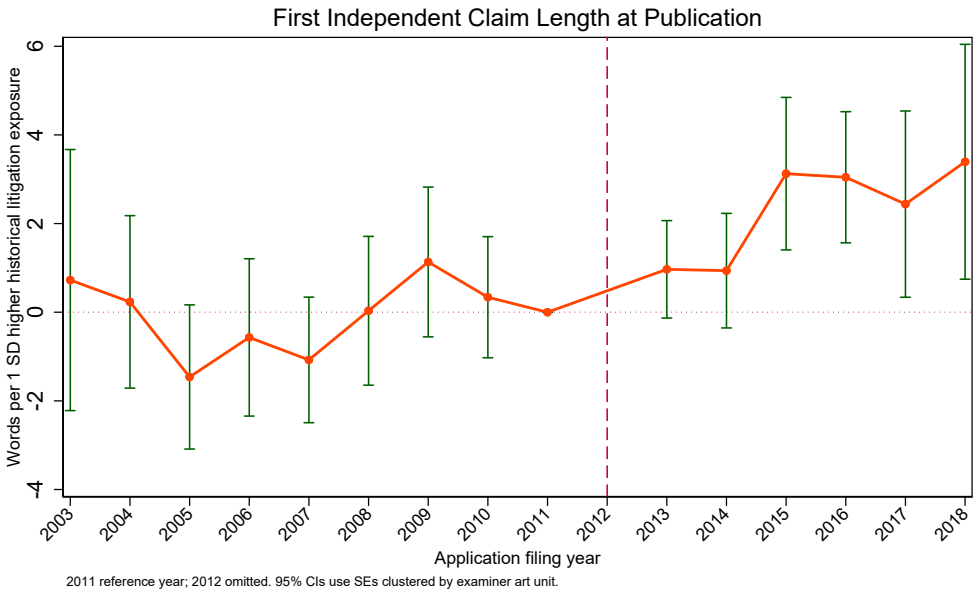


Figure 3: Historical Litigation Exposure and Published-Application Claim Length

Notes: The figure reports coefficients from interactions between standardized historical litigation exposure and application filing cohort. Examiner-art-unit and filing-cohort fixed effects are included. The 2011 filing cohort is the reference cohort and 2012 is excluded as a transition cohort. Regressions are weighted by the number of underlying applications in each examiner-art-unit-by-cohort cell. Vertical bars report 95 percent confidence intervals based on standard errors clustered by examiner art unit.

estimates are approximately 2.8 words in 2013 and 2014 and approximately 2.4 words in 2015 before declining to roughly one word in 2016. First independent claim length again produces a similar sequence.

The grant-stage results therefore contain some post-AIA evidence in the same direction as the publication results, but their interpretation is less straightforward. Substantial changes in the exposure gradient are already visible before the AIA, particularly between the earlier pre-period cohorts and 2010–2011. The post-AIA grant estimates cannot therefore be interpreted as a similarly clean break from a stable pre-period relationship. I return to these dynamics in Section 8, where I examine the role of weighting, art-unit composition, and matched publication–grant samples.

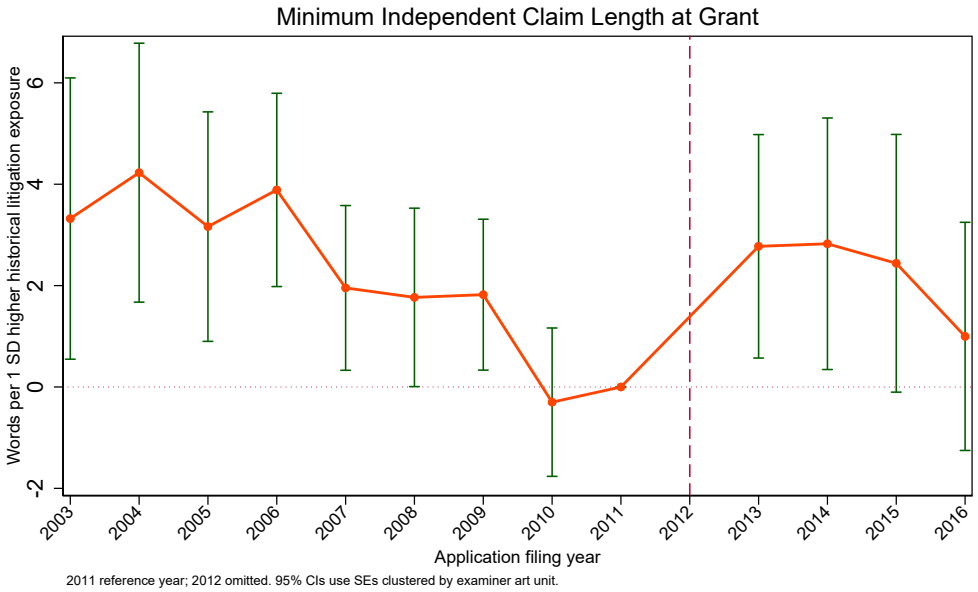
The publication-to-grant event studies provide additional information about how the exposure relationship evolves between the two stages. These estimates are based only on applications observed at both publication and grant. Claim-length changes are defined as grant-stage length minus publication-stage length, so larger positive values represent greater additional narrowing during prosecution.

For minimum independent claim length, the exposure gradient in prosecution-stage change is substantial in several early pre-AIA cohorts, reaching approximately three to four-and-a-half words relative to the 2011 gradient. It then declines through the pre-period and is close to zero by 2010 and 2011. First independent claim length displays the same basic pattern.

In the post-AIA period, the change gradients are initially positive in 2013 and 2014 but become negative in the later observed cohorts. For minimum independent claim length, the coefficient falls from approximately 1.6–1.8 words in 2013–2014 to approximately -0.9 words in 2015 and -2.3 words in 2016. The corresponding FICL estimates follow a similar path.

These dynamics make the prosecution-stage outcomes useful for understanding where the overall exposure relationship arises, but they also make a simple causal interpretation difficult. The decline in the prosecution-change gradient begins well before the AIA. I therefore treat these estimates primarily as stage-of-process evidence rather than as an independent difference-in-differences result. The full publication-to-grant event studies are reported in the appendix, and Section 8 examines the pattern using matched-sample decompositions and additional prosecution-stage diagnostics.

Panel A: Minimum Independent Claim Length



Panel B: First Independent Claim Length

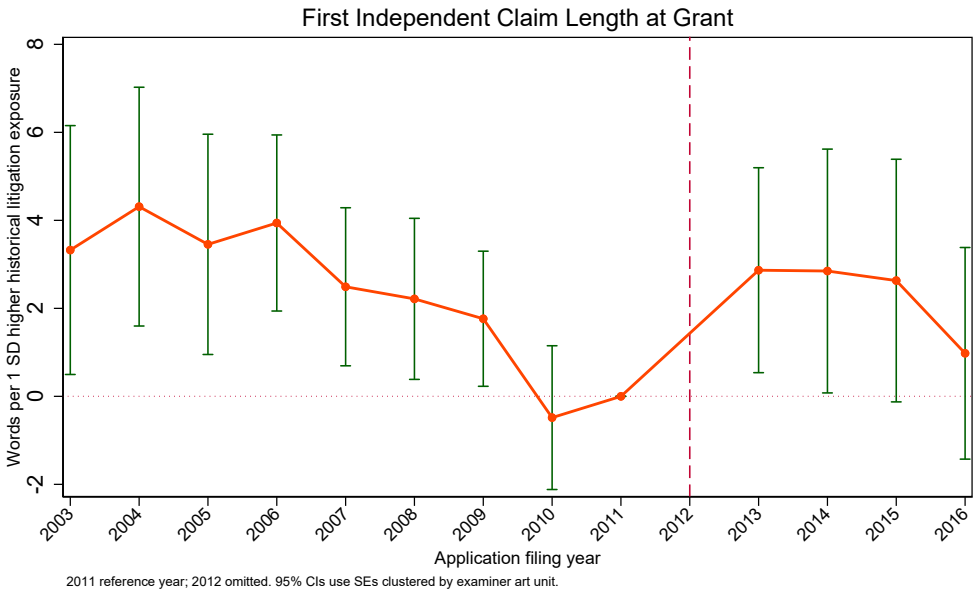


Figure 4: Historical Litigation Exposure and Granted Claim Length

Notes: The figure reports coefficients from interactions between standardized historical litigation exposure and application filing cohort for issued patents. Examiner-art-unit and filing-cohort fixed effects are included. The 2011 filing cohort is the reference cohort and 2012 is excluded as a transition cohort. Regressions are weighted by the number of underlying patents in each examiner-art-unit-by-cohort cell. Vertical bars report 95 percent confidence intervals based on standard errors clustered by examiner art unit.

7.2 Pooled Difference-in-Differences Estimates

Table 3 summarizes the event-study patterns using the pooled specification from Equation (1). Each coefficient measures the differential post-AIA change associated with a one-standard-deviation increase in historical litigation exposure.

Table 3: Historical Litigation Exposure and Patent Claim Outcomes

	Minimum ICL	First ICL	Independent Claim Count
Panel A: Published applications			
Exposure × Post	2.254***	2.326**	-0.0158
	(0.731)	(0.925)	(0.0126)
Underlying applications	4,164,106	4,156,739	4,167,068
Art-unit–cohort cells	6,271	6,271	6,271
Panel B: Patent grants			
Exposure × Post	0.296	0.256	0.0083
	(1.109)	(1.208)	(0.0105)
Underlying patents	2,596,100	2,596,089	2,596,109
Art-unit–cohort cells	5,463	5,463	5,463
Panel C: Publication-to-grant change			
Exposure × Post	-1.815*	-1.790*	0.0226
	(0.932)	(0.988)	(0.0144)
Underlying applications	2,401,596	2,397,378	2,403,286
Art-unit–cohort cells	5,462	5,461	5,462
Examiner-art-unit fixed effects	Yes	Yes	Yes
Filing-cohort fixed effects	Yes	Yes	Yes
Examiner art units	427	427	427

Notes: Each entry reports the coefficient on standardized historical litigation exposure interacted with an indicator for post-AIA filing cohorts. Historical exposure is constructed from district-court litigation involving patents granted during 2003–2011 and is grant-year adjusted before standardization. Filing cohorts 2003–2011 form the pre-period, the 2012 cohort is excluded as a transition cohort, and the post-period begins in 2013. Publication outcomes use filing cohorts through 2018; grant and publication-to-grant outcomes use cohorts through 2016. Regressions are estimated on examiner-art-unit-by-filing-cohort cells and weighted by the number of underlying observations. Standard errors, reported in parentheses, are clustered by four-digit examiner art unit. Panel C is restricted to applications observed at both publication and grant. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The publication estimates provide the paper's primary pooled results. A one-standard-deviation increase in historical litigation exposure is associated with a 2.254-word increase in minimum independent claim length at publication after the AIA, relative to the pre-AIA exposure relationship. The clustered standard error is 0.731, and the estimate is statistically significant at the one-percent level. Relative to the publication-stage mean of approximately 112 words, the estimate corresponds to an increase of roughly two percent.

The first-independent-claim estimate is very similar. A one-standard-deviation increase in historical exposure is associated with a 2.326-word increase in first independent claim length, with a

clustered standard error of 0.925. The similarity of the two estimates reinforces the event-study evidence that the post-AIA publication result is not specific to the particular construction of minimum independent claim length. In both cases, longer claims imply additional limitations and therefore relatively narrower initially requested breadth.

The publication independent-claim-count coefficient is -0.0158 claims, with a standard error of 0.0126. Although the pooled estimate is not statistically distinguishable from zero at conventional levels, its sign is consistent with the negative post-AIA event-study coefficients. I therefore interpret independent claim count as complementary directional evidence rather than as a separate headline result.

The pooled grant-stage estimates are much smaller. The coefficients for minimum and first independent claim length are 0.296 and 0.256 words, respectively, and neither is statistically distinguishable from zero. The grant results are therefore consistent with the event-study evidence that the strong publication-stage shift does not carry through to the population of issued claims as a comparably clear pooled change. Given the substantial pre-AIA movement in the grant exposure gradient, I do not interpret the near-zero pooled estimates as evidence that historical exposure is unrelated to granted claim breadth. Rather, the grant-stage relationship is already present and changing before the AIA, making the simple pre-versus-post summary less informative at that stage.

Finally, the pooled publication-to-grant coefficients are negative for both claim-length outcomes. A one-standard-deviation increase in historical exposure is associated with a 1.815-word smaller increase in minimum independent claim length between publication and grant and a 1.790-word smaller increase in first independent claim length. The corresponding p-values are approximately 0.052 and 0.071. These estimates are consistent with less additional claim narrowing during prosecution in more exposed art units in the post-AIA period, but the event-study evidence shows that the prosecution-change gradient was already declining before the AIA. I therefore do not interpret these coefficients as clean causal effects of the reform.

Taken together, the results point most clearly to a change in claims as they are initially presented by applicants. Historically more litigation-exposed examiner art units experience a larger post-AIA increase in both minimum and first independent claim length at publication. The corresponding grant-stage relationship is less clean because substantial exposure-related dynamics are already present before the reform, while the prosecution-stage evidence suggests that the additional narrowing occurring after publication becomes smaller in later cohorts. The next section examines

these stage-specific patterns and the sensitivity of the headline publication result in greater detail.

8 Robustness and Additional Evidence

The main results use examiner art units with at least 500 historical grants, impose no requirement that an art unit appear in every year of the historical period, and model historical litigation exposure continuously. This section examines whether the publication-stage results depend on those choices and provides additional evidence on how the exposure relationship evolves between publication and grant. I first vary the examiner-art-unit sample, then examine the support and functional form of historical exposure, and finally use a matched publication–grant sample to decompose the grant-stage exposure relationship into its publication-stage and prosecution-stage components.

8.1 Alternative Examiner-Art-Unit Samples

The preferred sample requires at least 500 eligible historical grants in an examiner art unit. This restriction is intended to limit noise in the historical litigation exposure measure without imposing a balanced-panel requirement on the historical patent data. I examine the sensitivity of the publication results to both features of this sample definition.

Table 4 first varies the minimum number of historical grants used to define the examiner-art-unit sample. To make the coefficients directly comparable, historical exposure is placed on the scale of the preferred 500-grant sample in every specification. The first row includes all usable four-digit examiner art units. The remaining unbalanced specifications successively require at least 250, 500, or 1,000 historical grants. I also report corresponding balanced specifications that require an art unit to be observed in every year from 2003 through 2011.

The results are stable across these alternatives. For minimum independent claim length, the estimated Exposure $\times$ Post coefficient ranges from 1.81 words when all four-digit art units are retained to 2.85 words under the 250-grant balanced restriction. The preferred estimate of 2.25 words lies near the middle of this range. First independent claim length shows a similar pattern, with estimates ranging from approximately 1.93 to 3.27 words. All of the alternative estimates remain positive, and their magnitudes are similar to the preferred specification despite substantial changes in the number of included examiner art units.

A separate exercise holds the 500-grant threshold fixed and progressively tightens the historical-

Table 4: Robustness to Examiner-Art-Unit Sample Definition

Sample definition	Minimum ICL	First ICL	Art units
All four-digit art units	1.807***	1.927***	683
	(0.549)	(0.683)	
250+ grants, unbalanced	2.411***	2.534***	473
	(0.635)	(0.798)	
500+ grants, unbalanced	2.255***	2.327**	427
	(0.731)	(0.925)	
1,000+ grants, unbalanced	2.189***	2.364**	350
	(0.799)	(0.933)	
250+ grants, balanced	2.849***	3.268***	252
	(0.749)	(0.918)	
500+ grants, balanced	2.699***	3.061***	240
	(0.873)	(1.066)	
1,000+ grants, balanced	2.168**	2.447**	217
	(0.936)	(1.121)	

Notes: Each entry reports the coefficient on standardized historical litigation exposure interacted with the post-AIA indicator. Exposure in all specifications is standardized using the preferred 500+ unbalanced examiner-art-unit sample. All regressions include examiner-art-unit and application-filing-year fixed effects, weight art-unit–filing-year cells by the number of underlying applications, omit the 2012 transition cohort, and cluster standard errors by examiner art unit. Balanced samples require an examiner art unit to be observed in each historical year from 2003 through 2011. Standard errors are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

coverage requirement. The preferred sample contains 427 examiner art units. Requiring an art unit to appear in at least six, seven, or eight of the nine historical years reduces the sample to 372, 335, and 297 art units, respectively, while requiring presence in all nine years leaves 240 art units. The corresponding publication sample falls from approximately 4.16 million applications to 2.88 million under the strictest requirement.

Figure 5 compares the resulting minimum-ICL event-study paths. The positive post-AIA pattern is present throughout the sequence of historical-coverage restrictions. Tightening the historical-presence requirement also reduces some of the irregularity in the pre-AIA coefficients, but at the cost of excluding a substantial share of the preferred publication sample. I therefore retain the 500-grant sample without an annual-presence requirement as the baseline rather than conditioning the analysis on continuous historical observation.

8.2 Exposure Support and Nonlinearity

Historical litigation exposure is right-skewed, raising the possibility that the continuous-exposure estimates are disproportionately influenced by a small number of highly exposed examiner art units. I examine this possibility in two ways. First, I re-estimate the preferred specification after excluding

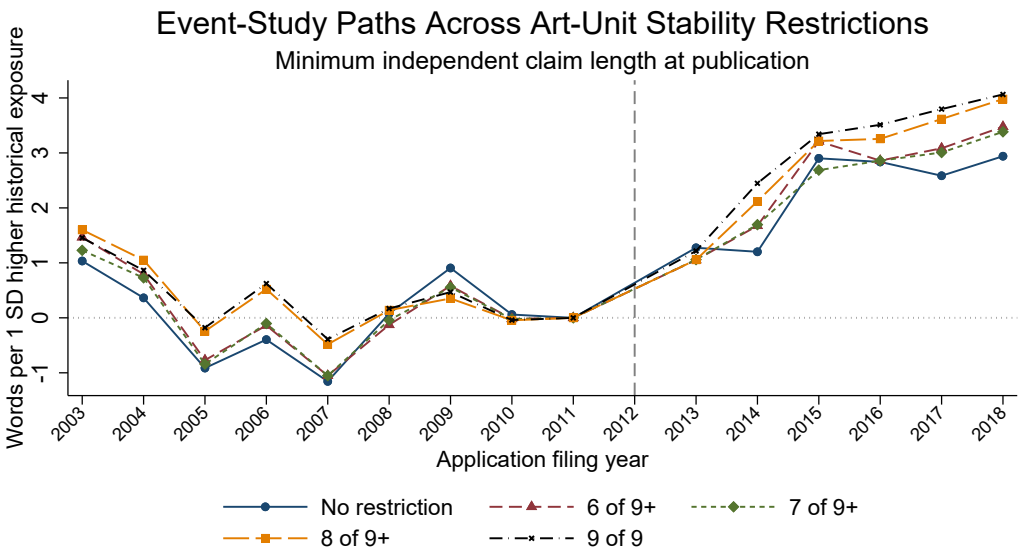


Figure 5: Event-Study Paths Across Historical Art-Unit Restrictions

Notes: The figure reports publication minimum-ICL event-study coefficients under progressively stricter requirements on the number of years in which an examiner art unit is observed during 2003–2011. All specifications require at least 500 historical grants. The 2011 filing cohort is the reference year and 2012 is omitted as the transition cohort. The figure is intended to compare the dynamic paths across sample definitions.

examiner art units above the 95th and 90th percentiles of the fixed historical exposure distribution. Second, I replace the continuous interaction with categorical exposure groups to examine the functional form of the post-AIA relationship.

Table 5 reports the tail-exclusion results. The exposure measure is not re-standardized after observations are removed, so all three estimates retain the original preferred-sample one-standard-deviation scale. For minimum independent claim length, the coefficient is 2.25 words in the preferred sample, 2.40 words after excluding the top 5 percent of examiner art units, and 2.48 words after excluding the top 10 percent. The point estimate therefore does not decline when the extreme upper tail is removed. The corresponding FICL estimates also remain positive, although they become less precisely estimated in the restricted samples.

Table 5: Robustness to the Upper Tail of Historical Litigation Exposure

	Preferred sample	Exclude top 5%	Exclude top 10%
Minimum ICL	2.254***	2.396**	2.479***
	(0.731)	(0.955)	(0.903)
First ICL	2.326**	2.252*	1.992*
	(0.925)	(1.219)	(1.119)

Notes: The table reports Exposure × Post estimates for publication claim-length outcomes. The upper-tail groups are defined from the fixed cross-sectional distribution of historical exposure across preferred examiner art units. Ties at the 90th and 95th percentile cutoffs are assigned to the upper group. Exposure remains standardized using the full preferred-sample scale after art units are excluded. All specifications include examiner-art-unit and filing-year fixed effects, application-count weights, and standard errors clustered by examiner art unit. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

The dynamic results reinforce this conclusion. Figure 6 overlays the minimum-ICL event study in the preferred sample with the corresponding paths after excluding the upper 5 and 10 percent of exposure. The positive coefficients in the later post-AIA cohorts remain after either exclusion. In addition, the pre-AIA path becomes noticeably less irregular when the upper decile is removed. This suggests that highly exposed examiner art units contribute to some of the earlier cross-cohort variation, but they are not necessary for the post-AIA publication pattern.

I do not adopt the restricted sample as the preferred specification. The tail exclusions were introduced as diagnostics of the support of the continuous exposure measure, and removing the most exposed technologies changes the population for which the relationship is estimated. The relevant robustness result is therefore that the publication finding survives the exclusions rather than that the restricted specification should replace the full preferred sample.

The categorical specifications provide a complementary check on linearity. For minimum ICL, the

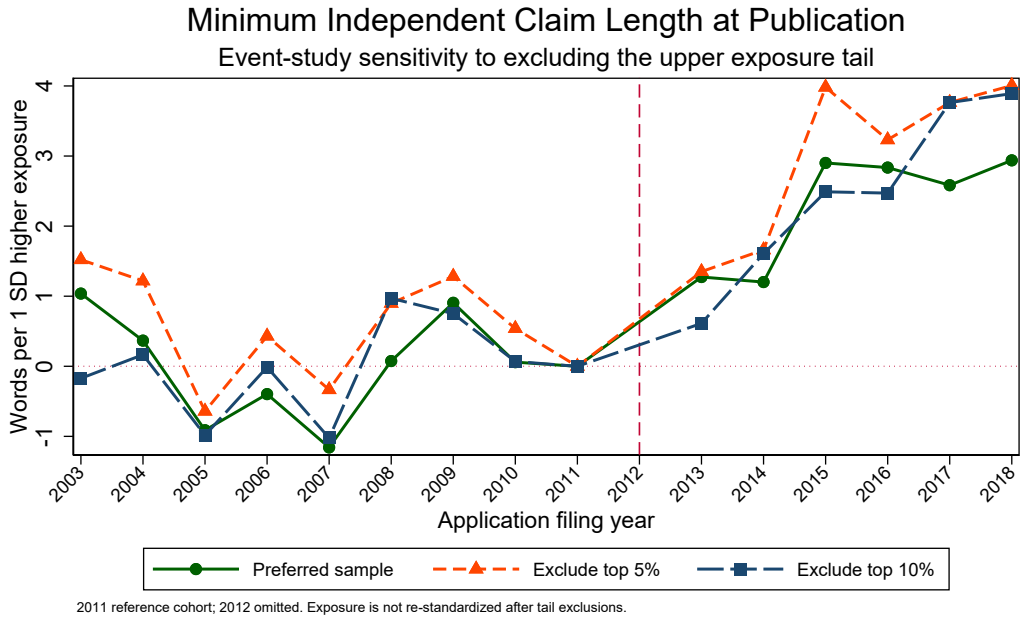


Figure 6: Event-Study Sensitivity to Excluding the Upper Exposure Tail

Notes: The figure compares publication minimum-ICL event-study point estimates for the preferred sample with estimates excluding the upper 5 and 10 percent of examiner art units by historical litigation exposure. Exposure is not re-standardized after the exclusions. The 2011 filing cohort is the reference year and 2012 is omitted.

estimated post-AIA difference relative to the lowest exposure quintile increases from approximately 0.36 words in the second quintile to 2.35, 3.50, and 4.74 words in the third through fifth quintiles. The upper two quintiles are estimated relatively precisely. The FICL coefficients are less monotonic, however, and direct comparisons of the top 5 or top 10 percent with the remainder of the distribution are imprecisely estimated. Taken together, these results suggest that the continuous specification is not masking a result generated solely by a few extreme examiner art units, while providing little reason to impose a particular nonlinear functional form. I therefore retain standardized continuous exposure as the preferred measure and report the full categorical estimates in the Appendix.

8.3 Matched-Sample Evidence and Prosecution Dynamics

The unrestricted publication and grant samples contain different sets of patent applications, so differences between their exposure gradients cannot be interpreted mechanically as changes occurring during prosecution. To examine the evolution of claim breadth within the same applications, I construct a matched sample containing applications for which minimum independent claim length

is observed at both publication and grant. For each application, the change during prosecution is defined as grant-stage minimum ICL minus publication-stage minimum ICL. The same applications and cell weights are then used for all three outcomes.

This construction produces an exact decomposition,

$$\beta_t^{Grant} = \beta_t^{Publication} + \beta_t^{Change}, \tag{3}$$

where each β_t is the filing-cohort-specific raw exposure gradient estimated on the common matched sample. The decomposition is an accounting relationship within this sample rather than an additional identification assumption.

Figure 7 shows the resulting components. In the early filing cohorts, the exposure gradient in minimum ICL at publication is close to zero, while the publication-to-grant change gradient is large and positive. The positive grant-stage exposure gradient in these cohorts is therefore generated primarily during the prosecution process. The prosecution component declines substantially over time. By the 2015 and 2016 filing cohorts, a positive exposure gradient is already visible when claims are published, while the additional gradient generated between publication and grant is much smaller.

The matched-sample event studies provide a second view of the same pattern. Publication-stage minimum ICL exhibits relatively little systematic pre-AIA exposure movement in the matched sample, followed by larger positive coefficients in 2015 and 2016. In contrast, the prosecution-change event study contains substantial positive exposure differences in several early cohorts relative to 2011 before declining toward the end of the pre-period. The grant-stage event study inherits these earlier prosecution dynamics. The corresponding FICL event studies show a similar division between a comparatively stable publication-stage pre-period and more substantial pre-existing prosecution-stage movement.

These results help explain why the grant-stage estimates receive a more cautious interpretation than the publication results. The positive historical relationship between litigation exposure and grant-stage claim length was already present before the AIA and appears to have been generated largely through prosecution. After the AIA, more of the exposure relationship is visible at publication, while the additional claim-length adjustment associated with exposure during prosecution falls. This pattern is consistent with a shift in the stage at which claim breadth responds to downstream

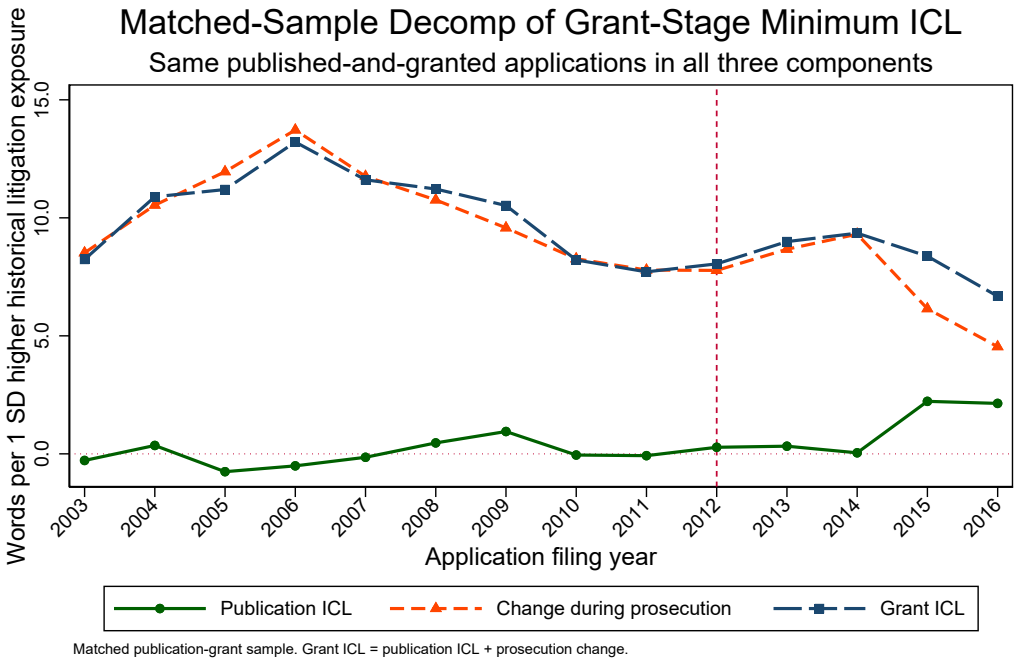


Figure 7: Matched-Sample Decomposition of Grant-Stage Minimum Independent Claim Length

Notes: The figure reports filing-cohort-specific historical-exposure gradients for publication minimum ICL, the change in minimum ICL between publication and grant, and grant minimum ICL. All three series use the identical matched publication–grant sample and identical art-unit–filing-year weights. By construction, the grant-stage gradient equals the publication-stage gradient plus the prosecution-change gradient in every filing cohort. The vertical line denotes the 2012 transition cohort.

challenge incentives.

The prosecution-change measure should not be interpreted as a direct measure of examiner effort. Changes between publication and grant reflect the equilibrium prosecution process, including applicant amendments, examiner objections, and other interactions between the applicant and the Patent Office. The matched decomposition therefore provides evidence about when differences in final claim breadth emerge, rather than assigning those changes exclusively to one side of the examination process.

Taken together, the robustness exercises show that the publication-stage results are not specific to the preferred examiner-art-unit cutoff or historical-coverage requirement, are not generated solely by the upper tail of the exposure distribution, and remain visible under more flexible representations of historical exposure. The matched evidence further shows that the relationship between historical exposure and claim breadth differs across stages of the patenting process. These exercises address several observable sources of specification and sample sensitivity but do not rule out every contemporaneous change that could differentially affect technologies with different historical litigation exposure. I return to those broader identification limitations in Section 9.

9 Discussion

The results point to a particular margin through which changes in the post-grant challenge environment can affect patenting behavior. The strongest evidence appears in the independent claims submitted by applicants at the publication stage, while the matched publication–grant evidence suggests that the stage at which claim breadth adjusts also changes across the sample period. This section first summarizes the empirical margin of response, then relates the findings to the strategic-search mechanism developed in the companion theoretical paper, and finally discusses the principal limitations of the empirical analysis.

9.1 The Margin of Response

The primary empirical response appears in the breadth of the independent claims entering the patenting process. More historically exposed examiner art units experience a larger post-AIA increase in both minimum and first independent claim length at publication. Because additional words and limitations in an independent claim generally restrict the set of subject matter covered by the

literal claim language, I interpret this pattern as evidence of relatively narrower initial claims in technological environments where downstream patent disputes had historically been more salient. By contrast, independent claim count does not exhibit a comparably strong pooled response. The principal margin therefore appears to be the content and breadth of independent claims rather than a simple change in the number of independent claims included in an application.

The stage-specific evidence suggests that the relevant margin is not only how much claim breadth changes, but also when that adjustment occurs. Before the AIA, relatively little systematic exposure gradient is visible in claim length at publication, while a substantially larger relationship is present by the time patents are granted. The matched publication–grant analysis shows that much of this historical grant-stage relationship emerges during prosecution. In later filing cohorts, however, a larger positive exposure gradient is already visible when applications are published, while the additional publication-to-grant exposure gradient is smaller. The evidence is therefore consistent with part of the adjustment in claim breadth moving upstream, from prosecution toward the applicant's initial claiming decision.

This distinction also helps reconcile the publication and grant results. The absence of a comparably strong pooled post-AIA coefficient at grant does not imply that the publication response disappears. Grant-stage claim breadth combines the claims with which applicants enter the process and all subsequent adjustment that occurs during prosecution. If applicants in more exposed technologies begin with relatively narrower claims, there may be less additional exposure-related adjustment left to occur before grant. The matched decomposition is consistent with such a reallocation across stages.

The publication-to-grant change measure should not, however, be interpreted as a direct measure of examiner behavior. Changes during prosecution reflect the equilibrium interaction between applicants and the Patent Office, including applicant amendments, examiner objections, responses to prior art, and other features of prosecution. The matched evidence is therefore most informative about the stage at which differences in final claim breadth emerge, rather than about which party alone generates those differences.

9.2 Connection to Strategic Search

These empirical patterns closely mirror the central mechanism of the strategic-search model developed in [3]. In that model, the applicant's search and claiming decisions depend not only on the

immediate examination process but also on the expected consequences of a later challenge. A change in the downstream challenger's incentives can therefore feed backward through the sequence of play and alter applicant behavior before the examiner has acted.

The post-grant-review provisions of the AIA provide a natural empirical setting for this mechanism. In the model, the closest analogue is a reduction in the cost of initiating a downstream challenge. Historical litigation exposure is not itself a structural primitive of the model, nor do I interpret more-exposed technologies as receiving a mechanically larger policy treatment. Rather, historical exposure provides a reduced-form measure of pre-existing differences in the economic relevance of downstream patent disputes. Technologies in which patents were historically more likely to be litigated may differ in the value of a successful challenge, the stakes associated with patent breadth, or other features that affect the profitability of downstream dispute.

This distinction provides a natural connection between the empirical exposure design and the theoretical model. In the model, the challenger files when its realized filing cost lies below a threshold $\widehat{K}_C$, where the threshold depends on the expected value of a challenge. Technological environments can therefore differ in $\widehat{K}_C$ even when they face the same distribution of challenge costs. Historical litigation exposure can be viewed as a reduced-form proxy for this type of pre-existing heterogeneity rather than as a direct measure of any individual model primitive.

A common reduction in challenge costs can then have different consequences across technologies without requiring the policy itself to differ across them. To see this, suppose that before the reform the filing probability in technological environment a is

$$p_{f,a}^{pre} = G_C^{pre}\left(\widehat{K}_{C,a}\right), \tag{4}$$

where G_C^{pre} is the pre-reform distribution of challenger filing costs. If the reform lowers challenge costs by a common amount $\Delta > 0$, the post-reform filing probability can be represented schematically as

$$p_{f,a}^{post} = G_C^{pre}\left(\widehat{K}_{C,a} + \Delta\right). \tag{5}$$

The resulting change in filing probability is

$$\Delta p_{f,a} = G_C^{pre}\left(\widehat{K}_{C,a} + \Delta\right) - G_C^{pre}\left(\widehat{K}_{C,a}\right). \tag{6}$$

Even though Δ is common across technologies, the behavioral effect need not be identical because technological environments can begin with different challenge thresholds. The empirical interaction between historical exposure and the post-AIA period is therefore best viewed as asking whether a common change in the challenge environment had different consequences across settings in which downstream patent disputes had previously differed in economic importance.

This framework also helps interpret the weak relationship between historical exposure and publication-stage claim length before the AIA. If pre-reform challenge costs were sufficiently high, filing probabilities could have been small across a wide range of technological environments. In that region, even meaningful differences in the value or profitability of downstream challenge would translate into only small differences in expected challenge risk. Those differences would therefore have relatively little effect on the applicant's expected return from entering with broader claims and little reason to generate a strong exposure gradient in initial claim drafting.

A sufficiently large common reduction in challenge costs can change that calculus. Once challenge becomes a more meaningful possibility, pre-existing differences in the downstream environment can generate larger differences in expected challenge risk and therefore larger differences in the return to the applicant's initial claiming decision. The post-AIA emergence of a publication-stage exposure gradient is thus consistent with the reform making previously existing heterogeneity in downstream challenge incentives more behaviorally relevant, rather than with applicants only beginning to recognize those differences after the AIA.[2]

The stage-specific evidence provides a second connection to the model. Search and adjustment occur sequentially, so behavior by an earlier player changes the claims and information inherited by later players. If a stronger downstream challenge environment gives applicants greater incentive to search, draft more carefully, or begin prosecution with narrower claims, the examiner inherits a different starting point. The marginal need for additional narrowing later in prosecution can then decline.

The matched-sample results are consistent with this type of interdependence across stages. Historically, relatively little of the exposure relationship is visible when applications are published, while a substantial portion emerges between publication and grant. In later post-AIA cohorts, more of

[2]This interpretation describes one parameter region of the theoretical model rather than a separate structural result established by the empirical analysis. Whether a common change in filing costs magnifies cross-sectional differences in challenge probabilities depends on the location of the relevant filing thresholds within the challenger cost distribution. Historical litigation exposure also need not map one-to-one into any single model primitive.

the relationship is already present in the applicant's published claims and less additional exposure-related adjustment occurs during prosecution. In this sense, the relevant margin appears to move upstream. A common change in downstream challenge incentives makes pre-existing differences in challenge stakes more consequential for the applicant's initial decision, changing the claims that enter examination and therefore the starting point inherited by later stages of prosecution.

The empirical evidence should not be interpreted as a structural test of each comparative static in the theoretical model. I do not directly observe applicant, examiner, or challenger search effort, and the publication-to-grant change measure does not isolate examiner actions. Historical litigation exposure is likewise a reduced-form measure of the downstream environment rather than an estimate of a structural parameter. What the evidence provides is support for the broader strategic relationship at the center of the model: changes in downstream challenge incentives are associated with changes in upstream applicant behavior, and adjustment at one stage of the patenting process is related to the amount of adjustment that subsequently occurs at later stages.

The grant-stage evidence provides a further, although more tentative, connection to the substitution mechanism in the model. The grant event studies do not support a clean causal interpretation because the relationship between historical exposure and grant-stage claim length is already evolving before the AIA. Nevertheless, the matched decomposition is suggestive about how the stages interact. In the later post-AIA cohorts, the positive exposure relationship that emerges at publication is accompanied by a smaller, and eventually negative, exposure relationship in the additional change that occurs during prosecution. For example, in the 2015 and 2016 filing cohorts, the matched publication-stage exposure coefficients for minimum independent claim length are approximately 2.8 words, while the corresponding publication-to-grant change coefficients are approximately -0.9 and -2.3 words. The resulting grant-stage coefficients are therefore much smaller, at approximately 1.9 and 0.5 words.

This pattern is consistent with partial substitution across stages. If applicants in more exposed technologies enter prosecution with narrower claims, less additional exposure-related narrowing may be required later in the patenting process. The result is not that the initial response necessarily disappears by grant, but that part of it is offset by a smaller subsequent adjustment. Because prosecution-stage changes reflect both applicant and examiner behavior, this evidence cannot identify examiner substitution directly. It nevertheless provides a reduced-form pattern consistent with the model's prediction that stronger adjustment by an earlier player can reduce the marginal need for

adjustment by later players.

9.3 Limitations

Several limitations qualify the interpretation of these results. First, the empirical design is reduced form. Historical litigation exposure is predetermined relative to the post-AIA filing cohorts, and examiner-art-unit fixed effects absorb persistent differences across technological examination environments, but historical exposure is not randomly assigned. The identifying assumption is that, absent the AIA, technologies with different historical litigation exposure would not have experienced systematically different changes in publication-stage claim breadth coinciding with the reform.

The publication event studies and the robustness exercises provide evidence relevant to this assumption. The publication coefficients do not display a sustained pre-AIA movement in the same direction as the later post-AIA pattern, and the main result is robust to alternative examiner-art-unit samples and to excluding the upper tail of the historical exposure distribution. Nevertheless, the design cannot rule out every contemporaneous technology-specific shock that may have been correlated with historical litigation exposure. The estimates should therefore be interpreted as differential changes associated with historical exposure around the AIA rather than as a fully isolated structural treatment effect.

Second, the AIA changed several features of the patent system. The theoretical interpretation emphasizes the post-grant challenge provisions, particularly the creation of inter partes review, because they provide the most direct connection to the challenger stage of the model. The empirical design, however, identifies the differential response to the broader post-AIA institutional environment. It does not separately identify the effect of lower challenge costs from every other provision of the AIA or from other contemporaneous institutional changes that may have affected technologies differently.

For the same reason, historical litigation exposure should not be interpreted as a direct measure of the treatment generated by the AIA. It summarizes realized pre-AIA litigation patterns and therefore reflects the equilibrium consequences of several underlying features, including patent values, dispute stakes, technological characteristics, and the incentives of potential challengers. Its usefulness in the empirical design comes from capturing persistent differences in the historical importance of downstream patent disputes, not from identifying the structural source of those differences.

Third, independent claim length is an informative but imperfect measure of patent breadth.

Adding words or limitations to an independent claim generally restricts the set of subject matter covered by its literal language, but legal breadth also depends on the substantive content and interaction of those limitations, claim construction, the specification, the relevant prior art, and subsequent legal interpretation. The similar results for minimum and first independent claim length, together with the claim reconstruction and validation exercises, reduce concern that the findings are particular to one arbitrary textual measure, but neither measure provides a complete representation of economic or legal patent breadth.

Finally, outcomes observed later in prosecution are more difficult to interpret causally than publication-stage outcomes. Applications must survive long enough to be granted to enter the grant sample, the matched sample necessarily conditions on eventual issuance, and recent filing cohorts are more exposed to right-censoring in grant data ending in 2020. More importantly, both the grant-stage and publication-to-grant-change event studies exhibit meaningful exposure-related dynamics before the AIA. These patterns indicate that the relationship between historical exposure and later-stage patent outcomes was already evolving before the reform.

For these reasons, I treat the publication-stage results as the primary evidence on the response to the AIA. The grant and matched-sample results play a different role. They help show how the relationship between downstream challenge exposure and eventual claim breadth is distributed across stages of the patenting process and provide evidence that, following the reform, a larger share of that relationship appears earlier in the applicant's claiming decision.

10 Conclusion

This paper examines whether changes in the incentives for downstream patent challenge affect behavior earlier in the patenting process. I combine newly reconstructed claims data with historical patent-litigation exposure across examiner art units and use the America Invents Act as a common change in the post-grant challenge environment. Following the AIA, applications in more historically exposed art units enter the patenting process with relatively longer, and therefore narrower, independent claims. In the preferred specification, a one-standard- deviation increase in historical litigation exposure is associated with approximately 2.25 additional words in minimum independent claim length at publication. A similar relationship appears for first independent claim length, and the publication results are robust to alternative examiner-art-unit samples and to changes in the

support and functional form of historical exposure.

The stage-specific evidence further suggests that the response involved a change in when claim-breadth adjustment occurred. Before the AIA, relatively little of the historical exposure relationship is visible at publication, while a substantial portion emerges during prosecution. In later post-AIA cohorts, more of that relationship is already present in the applicant's published claims and less additional exposure-related adjustment occurs between publication and grant. The grant-stage results do not provide the same clean causal evidence because similar exposure gradients were already evolving before the reform. Nevertheless, the matched publication–grant decomposition is consistent with partial substitution across stages, as greater adjustment before examination is accompanied by less additional adjustment during prosecution.

These findings provide empirical support for the broader strategic relationship developed in the companion theoretical model. Downstream challenge incentives need not affect only the decision to challenge a granted patent. By changing the expected consequences of the claims that an applicant brings into the patenting process, they can alter behavior well before a challenge occurs. The results therefore suggest that institutions governing post-grant disputes can influence not only which patents are ultimately challenged, but also the breadth of the patent rights applicants initially seek and the stage of the patenting process at which those rights are shaped.

References

[1] Talia Bar and Brendan Costello. "Predicting Administrative Patent Challenges". In: *Minn. J. Law Sci. Technol.* 21.1 (2020), p. 207.

[2] Rudi Bekkers, Arianna Martinelli, and Federico Tamagni. "The impact of including standards-related documentation in patent prior art: Evidence from an EPO policy change". In: *Research Policy* 49.7 (2020). DOI: 10.1016/j.respol.2020.104007. URL: https://ideas.repec.org/a/eee/respol/v49y2020i7s004873332030086x.html.

[3] Nicholas Dunagan. "Strategic Search in Patenting". Working paper. 2026.

[4] Josh Feng and Xavier Jaravel. "Crafting Intellectual Property Rights: Implications for Patent Assertion Entities, Litigation, and Innovation". In: *American Economic Journal: Applied Economics* 12.1 (Jan. 2020), pp. 140–181. DOI: 10.1257/app.20180361. URL: https://www.aeaweb.org/articles?id=10.1257/app.20180361.

[5] Michael D. Frakes and Melissa F. Wasserman. "Is the Time Allocated to Review Patent Applications Inducing Examiners to Grant Invalid Patents?: Evidence from Micro-Level Application Data". In: *The Review of Economics and Statistics* 99.3 (2017), pp. 550–563. ISSN: 00346535, 15309142. URL: https://www.jstor.org/stable/26616139.

[6] Richard Gilbert and Carl Shapiro. "Optimal Patent Length and Breadth". In: *The RAND Journal of Economics* 21.1 (1990), pp. 106–112. ISSN: 07416261. URL: http://www.jstor.org/stable/2555497.

[7] Stuart J. H. Graham, Alan C. Marco, and Richard Miller. "The USPTO Patent Examination Research Dataset: A Window on Patent Processing". In: *Journal of Economics & Management Strategy* 27.3 (2018), pp. 554–578. DOI: 10.1111/jems.12263.

[8] Stuart J.H. Graham and Dietmar Harhoff. "Separating patent wheat from chaff: Would the US benefit from adopting patent post-grant review?" In: *Research Policy* 43.9 (2014), pp. 1649–1659. ISSN: 0048-7333. DOI: https://doi.org/10.1016/j.respol.2014.07.002. URL: https://www.sciencedirect.com/science/article/pii/S0048733314001140.

[9] Bronwyn H. Hall et al. "Prospects for Improving U.S. Patent Quality via Postgrant Opposition". In: *Innovation Policy and the Economy* 4 (2004), pp. 115–143. DOI: 10.1086/ipe.

4.25056163. URL: https://www.journals.uchicago.edu/doi/abs/10.1086/ipe.4.25056163.

[10] Dietmar Harhoff and Markus Reitzig. "Determinants of opposition against EPO patent grants—the case of biotechnology and pharmaceuticals". In: *International Journal of Industrial Organization* 22.4 (2004), pp. 443–480. ISSN: 0167-7187. DOI: https://doi.org/10.1016/j.ijindorg.2004.01.001. URL: https://www.sciencedirect.com/science/article/pii/S0167718704000219.

[11] Deepak Hegde, Kyle Herkenhoff, and Chenqi Zhu. "Patent Publication and Innovation". In: *Journal of Political Economy* 131.7 (2023), pp. 1845–1903. DOI: 10.1086/723636. URL: https://www.journals.uchicago.edu/doi/abs/10.1086/723636.

[12] Deepak Hegde, David C. Mowery, and Stuart J.H. Graham. "Pioneers, Submariners, or Thicket-Builders: Which Firms Use Continuations in Patenting?" In: *Management Science* 55.7 (2009), pp. 1214–1226. DOI: 10.12887/mnsc.1090.1016.

[13] Paul Klemperer. "How Broad Should the Scope of Patent Protection Be?" In: *RAND Journal of Economics* 21.1 (1990), pp. 113–130. URL: https://ideas.repec.org/a/rje/randje/v21y1990ispringp113-130.html.

[14] Jeffrey M. Kuhn and Neil Thompson. "How to Measure and Draw Causal Inferences with Patent Scope". In: *SSRN* (2019). URL: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=2977273.

[15] Jean O. Lanjouw and Mark Schankerman. "Protecting Intellectual Property Rights: Are Small Firms Handicapped?*". In: *The Journal of Law and Economics* 47.1 (2004), pp. 45–74. DOI: 10.1086/380476. URL: https://www.journals.uchicago.edu/doi/abs/10.1086/380476.

[16] Mark A. Lemley and Bhaven Sampat. "Examiner Characteristics and Patent Office Outcomes". In: *The Review of Economics and Statistics* 94.3 (2012), pp. 817–827. ISSN: 00346535, 15309142. URL: http://www.jstor.org/stable/23261480.

[17] Mark A. Lemley and Carl Shapiro. "Probabilistic Patents". In: *Journal of Economic Perspectives* 19.2 (June 2005), pp. 75–98. DOI: 10.1257/0895330054048650. URL: https://www.aeaweb.org/articles?id=10.1257/0895330054048650.

[18] Joshua Lerner. "The Importance of Patent Scope: An Empirical Analysis". In: *The RAND Journal of Economics* 25.2 (1994), pp. 319–333. ISSN: 07416261. URL: http://www.jstor.org/stable/2555833.

[19] Brian J. Love and Shawn Ambwani. "Inter Partes Review: An Early Look at the Numbers". In: *The University of Chicago Law Review Dialogue* 81 (Oct. 2014). URL: https://chicagounbound.uchicago.edu/uclrev_online/vol81/iss1/5/.

[20] Brian J. Love, Shawn P. Miller, and Shawn Ambwani. "Determinants of Patent Quality: Evidence from Inter Partes Review Proceedings". In: *University of Colorado Law Review* 90 (2019), p. 67. URL: https://scholar.law.colorado.edu/lawreview/vol90/iss1/3/.

[21] Alan C. Marco, Joshua D. Sarnoff, and Charles A.W. deGrazia. "Patent claims and patent scope". In: *Research Policy* 48.9 (2019), p. 103790. ISSN: 0048-7333. DOI: https://doi.org/10.1016/j.respol.2019.04.014. URL: https://www.sciencedirect.com/science/article/pii/S0048733319301052.

[22] Arianna Martinelli and Julia Mazzei. "Death squad or quality improvement? The impact of introducing post-grant review on U.S. patent legal quality". In: *Research Policy* 54.5 (2025), p. 105205. ISSN: 0048-7333. DOI: 10.1016/j.respol.2025.105205. URL: https://www.sciencedirect.com/science/article/pii/S0048733325000344.

[23] Robert P. Merges and Richard R. Nelson. "On the Complex Economics of Patent Scope". In: *Columbia Law Review* 90.4 (1990), pp. 839–916. ISSN: 00101958. URL: http://www.jstor.org/stable/1122920.

[24] Richard D. Miller. *Technical Documentation for the 2019 Patent Examination Research Dataset (PatEx) Release*. USPTO Economic Working Paper 2020-4. United States Patent and Trademark Office, Office of the Chief Economist, Aug. 2020. DOI: 10.2139/ssrn.3675671. URL: https://ssrn.com/abstract=3675671.

[25] Markus Nagler and Stefan Sorg. "The disciplinary effect of post-grant review – Causal evidence from European patent opposition". In: *Research Policy* 49.3 (2020), p. 103915. ISSN: 0048-7333. DOI: https://doi.org/10.1016/j.respol.2019.103915. URL: https://www.sciencedirect.com/science/article/pii/S0048733319302331.

[26] Elena Novelli. "An examination of the antecedents and implications of patent scope". In: *Research Policy* 44.2 (2015), pp. 493–507. ISSN: 0048-7333. DOI: https://doi.org/10.1016/j.respol.2014.09.005. URL: https://www.sciencedirect.com/science/article/pii/S0048733314001619.

[27] Ted O'Donoghue, Suzanne Scotchmer, and Jacques-François Thisse. "Patent Breadth, Patent Life, and the Pace of Technological Progress". In: *Journal of Economics & Management Strategy* 7.1 (1998), pp. 1–32. DOI: https://doi.org/10.1111/j.1430-9134.1998.00001.x.

[28] Cesare Righi, Davide Cannito, and Theodor Vladasel. "Continuing patent applications at the USPTO". In: *Research Policy* 52.4 (2023), p. 104742. ISSN: 0048-7333. DOI: https://doi.org/10.1016/j.respol.2023.104742. URL: https://www.sciencedirect.com/science/article/pii/S0048733323000264.

[29] Cesare Righi and Timothy Simcoe. "Patent examiner specialization". In: *Research Policy* 48.1 (2019), pp. 137–148. ISSN: 0048-7333. DOI: https://doi.org/10.1016/j.respol.2018.08.003. URL: https://www.sciencedirect.com/science/article/pii/S0048733318301902.

[30] Mark Schankerman and Florian Schuett. "Patent Screening, Innovation, and Welfare". In: *The Review of Economic Studies* 89.4 (2021), pp. 2101–2148. ISSN: 0034-6527. DOI: 10.1093/restud/rdab073. URL: https://doi.org/10.1093/restud/rdab073.

[31] David L. Schwartz, Ted M. Sichelman, and Richard Miller. *USPTO Patent Number and Case Code File Dataset Documentation*. USPTO Economic Working Paper 2019-05. Also Northwestern Law & Economics Research Paper No. 19-12. United States Patent and Trademark Office, Office of the Chief Economist, Dec. 2019. DOI: 10.2139/ssrn.3507607. URL: https://ssrn.com/abstract=3507607.

[32] Suzanne Scotchmer. "Standing on the Shoulders of Giants: Cumulative Research and the Patent Law". In: *Journal of Economic Perspectives* 5.1 (Mar. 1991), pp. 29–41. DOI: 10.1257/jep.5.1.29. URL: https://www.aeaweb.org/articles?id=10.1257/jep.5.1.29.

[33] Andrew A. Toole, Richard Miller, and Ted M. Sichelman. *Technical Documentation for Patent Litigation Docket Reports Data, 1963–2020*. USPTO Economic Working Paper 2024-01. United States Patent and Trademark Office, Office of the Chief Economist, Mar. 2024. DOI: 10.2139/ssrn.4780166. URL: https://ssrn.com/abstract=4780166.

[34] United States Congress. *Leahy-Smith America Invents Act*. Pub. L. No. 112-29, 125 Stat. 284. Accessed September 26, 2026. 2011. URL: https://www.govinfo.gov/app/details/PLAW-112publ29.

[35] United States Patent and Trademark Office. *First Inventor to File (FITF) Resources*. Accessed September 26, 2026. 2026. URL: https://www.uspto.gov/patents/first-inventor-file-fitf-resources.

[36] United States Patent and Trademark Office. *Information Disclosure Statement (IDS) Resources*. Accessed March 23, 2026. 2026. URL: https://www.uspto.gov/patents/apply/filing-online/information-disclosure-statement-ids-resources.

[37] United States Patent and Trademark Office. *Inter Partes Disputes*. Accessed September 26, 2026. 2026. URL: https://www.uspto.gov/patents/laws/america-invents-act-aia/inter-partes-disputes.

[38] United States Patent and Trademark Office. *Inter Partes Review*. Accessed September 26, 2026. 2026. URL: https://www.uspto.gov/patents/ptab/trials/inter-partes-review.

[39] United States Patent and Trademark Office. *Manual of Patent Examining Procedure, Section 1120: Eighteen-Month Publication of Patent Applications*. MPEP Section 1120; Accessed September 26, 2026. 2026. URL: https://www.uspto.gov/web/offices/pac/mpep/s1120.html.

[40] United States Patent and Trademark Office. *Manual of Patent Examining Procedure, Section 1121: Content of a Patent Application Publication*. MPEP Section 1121; Accessed September 26, 2026. 2026. URL: https://www.uspto.gov/web/offices/pac/mpep/s1121.html.

[41] United States Patent and Trademark Office. *Manual of Patent Examining Procedure, Section 2001: Duty of Disclosure, Candor, and Good Faith*. MPEP Section 2001; Accessed March 23, 2026. 2026. URL: https://www.uspto.gov/web/offices/pac/mpep/s2001.html.

[42] United States Patent and Trademark Office. *Manual of Patent Examining Procedure, Section 2201: Introduction*. MPEP Section 2201; Accessed September 26, 2026. 2026. URL: https://www.uspto.gov/web/offices/pac/mpep/s2201.html.

[43] United States Patent and Trademark Office. *Manual of Patent Examining Procedure, Section 609: Information Disclosure Statement*. MPEP Section 609; Accessed September 26, 2026. 2026. URL: https://www.uspto.gov/web/offices/pac/mpep/s609.html.

[44] United States Patent and Trademark Office. *Manual of Patent Examining Procedure, Section 704: Nature of Examiner's Action*. MPEP Section 704; Accessed March 23, 2026. 2026. URL: https://www.uspto.gov/web/offices/pac/mpep/s704.html.

[45] United States Patent and Trademark Office. *Manual of Patent Examining Procedure, Section 706: Rejection of Claims*. MPEP Section 706; Accessed March 23, 2026. 2026. URL: https://www.uspto.gov/web/offices/pac/mpep/s706.html.

[46] United States Patent and Trademark Office. *Manual of Patent Examining Procedure, Section 713: Interviews*. MPEP Section 713; Accessed March 23, 2026. 2026. URL: https://www.uspto.gov/web/offices/pac/mpep/s713.html.

[47] United States Patent and Trademark Office. *Manual of Patent Examining Procedure, Section 714: Amendments, Applicant's Action*. MPEP Section 714; Accessed September 26, 2026. 2026. URL: https://www.uspto.gov/web/offices/pac/mpep/s714.html.

[48] United States Patent and Trademark Office. *Manual of Patent Examining Procedure, Section 904: How to Search*. MPEP Section 904; Accessed September 26, 2026. 2026. URL: https://www.uspto.gov/web/offices/pac/mpep/s904.html.

[49] United States Patent and Trademark Office. *Nonprovisional (Utility) Patent Application Filing Guide*. Accessed September 26, 2026. June 2023. URL: https://www.uspto.gov/patents/basics/apply/utility-patent.

[50] United States Patent and Trademark Office. *Patent Process Overview*. Accessed March 23, 2026. 2026. URL: https://www.uspto.gov/patents/basics/patent-process-overview.

[51] United States Patent and Trademark Office. *Patents Examination*. Accessed September 26, 2026. 2026. URL: https://www.uspto.gov/patents/laws/america-invents-act-aia/patents-examination.

[52] United States Patent and Trademark Office. *Post Grant Review*. Accessed September 26, 2026. 2026. URL: https://www.uspto.gov/patents/ptab/trials/post-grant-review.

A Claims Construction and Validation

The empirical analysis relies on claim measures reconstructed from the USPTO full-text files beyond the period covered by the historical USPTO Patent Claims Research Dataset. I therefore validate the reconstruction against the historical data in 2014, the final year for which the two sources overlap. I conduct the exercise separately for published applications and issued patents because the historical application and grant files differ in their identifiers and time coverage.

The validation proceeds at both the claim and document levels. At the claim level, I compare whether the reconstruction recovers the same claims, whether each claim is classified as independent or dependent, and whether the reconstructed text statistics reproduce the historical word counts. For published applications, I additionally compare cancellation status. Because first independent claim length is used directly as an outcome in the empirical analysis, I also construct the first non-canceled independent claim separately from each dataset and compare both its claim number and its word count. At the document level, I compare the number of independent and dependent claims and the minimum, average, and total word counts constructed from those claims.

Table 6 reports the application-stage validation. Of the comparable historical claims, 6,725,663 are recovered by the reconstruction and only 46 are unmatched, corresponding to coverage above 99.999 percent. Independent-versus-dependent status agrees exactly for 99.99989 percent of comparable claims and cancellation status agrees for 99.99906 percent. Exact word-count agreement is somewhat lower because of small differences in historical tokenization conventions, but 99.99635 percent of matched non-canceled claims differ by no more than one word.

The additional first-independent-claim comparison produces similarly strong agreement. The reconstruction identifies the same first independent claim in 99.99811 percent of the 370,679 comparable publications. First independent claim length agrees exactly in 99.48122 percent of these documents and differs by no more than one word in 99.99002 percent. The difference between exact and within-one-word agreement mirrors the claim-level word-count results and reflects the same small tokenization differences rather than disagreement about which claim is the first independent claim.

The document-level comparison matches 379,680 publications across the two sources. Independent claim counts agree exactly for 99.07896 percent of matched documents and dependent claim counts for 98.76791 percent. These rates are lower than the underlying claim-level classification

agreement because the historical claim-level and document-level files are not mechanically consistent with one another. Reaggregating the historical claim-level records does not perfectly reproduce the historical document-level counts. Conditional on the two sources containing the same number of independent claims, however, the word-based measures used in this paper agree almost perfectly. Minimum independent claim length agrees exactly for 99.98671 percent of the 376,183 comparable documents, while the corresponding average and total word measures agree in more than 99.97 percent of cases.

Table 7 reports the corresponding validation for issued patents. The grant comparison differs from the application exercise because the historical grant files do not contain a usable grant-year variable. Reconstructed 2014 claims must therefore be matched against the full historical grant-claim universe using application number, patent number, and claim number. The common patent-number key is numeric, so this validation comparison is restricted to utility patents. Non-utility patents remain in the reconstructed data and are omitted only from the validation exercise. Historical-only observations cannot be interpreted as unmatched 2014 grants, so grant-side coverage is measured relative to the reconstructed 2014 utility-patent comparison universe.

Among 4,995,011 reconstructed utility-patent claims available for comparison, 4,978,864 are located in the historical data, corresponding to coverage of 99.67674 percent. Independent-versus-dependent classification agrees exactly for 99.99998 percent of matched claims, with only one classification disagreement. Word counts agree exactly for 99.98891 percent of matched claims and are within one word for 99.99827 percent.

The direct first-independent-claim validation is similarly strong. Among 300,101 comparable patents, the first independent claim number agrees in 99.99900 percent of cases. Its word count agrees exactly in 99.98967 percent and lies within one word of the historical measure in 99.99034 percent.

At the patent level, 300,671 of 301,643 reconstructed utility-patent documents match a historical patent record, a match rate of 99.67776 percent. Independent and dependent claim counts each agree exactly in 99.87927 percent of matched patents, while total claim count agrees in every matched patent. The small independent/dependent count differences therefore reflect offsetting differences in how a small number of claims are classified rather than missing patent claims. Conditional on agreement in the relevant claim count, minimum independent claim length agrees exactly in 99.98701 percent of patents and the corresponding average and total independent-claim word measures agree

in more than 99.98 percent of comparisons.

Taken together, the claim- and document-level exercises show that the reconstruction closely reproduces the historical USPTO measures on the dimensions used in the empirical analysis. The remaining discrepancies are concentrated in isolated classification differences, very small word-count differences, and differences in aggregation conventions between the historical claim- and document-level files. I therefore use the reconstructed measures to extend the historical claims data through 2020.

B Litigation Exposure Construction and Distribution

This appendix provides additional details on the construction and distribution of the historical litigation exposure measure used in the main analysis. As described in Section 5, exposure is constructed at the four-digit examiner-art-unit level using regular utility patents granted between 2003 and 2011. A patent is classified as historically litigated if it is observed in federal district-court litigation during this same period. [3]

For examiner art unit a, the raw historical litigation exposure measure is

$$E_a^{\text{raw}} = 1000 \left(\frac{\sum_{i \in a} L_i}{N_a} \right), \tag{7}$$

where L_i is an indicator equal to one if patent i was litigated during the historical period and N_a is the number of eligible grants in art unit a. The raw measure is therefore expressed as historically litigated patents per 1,000 grants.

Because patents granted in different years have different amounts of time in which litigation can be observed before the end of the historical window, I adjust for common grant-year differences in observed litigation propensity. Specifically, I estimate

$$L_i = \delta_{g(i)} + \varepsilon_i, \tag{8}$$

where $\delta_{g(i)}$ denotes grant-year fixed effects. I then average the residuals within examiner art unit and add the overall historical litigation rate back to obtain

[3] The historical exposure measure treats litigation filed throughout calendar year 2011 as part of the pre-AIA historical window. The AIA was enacted on September 16, 2011. I retain the calendar-year definition in the baseline construction.

Table 6: Validation of Reconstructed Published-Application Claims, 2014

Validation measure	Comparison N	Agreement / result
Panel A: Claim-level validation		
Historical claims matched	6,725,663	99.99932%
Historical claims unmatched	46	0.00068%
Reconstruction-only claims	1,344,454	—
Reconstruction-only claims canceled	1,132,593	84.24186%
Independent/dependent classification	6,625,194	99.99989% exact
Cancellation status	6,725,663	99.99906% exact
Word count	6,625,231	99.38773% exact
Word count	6,625,231	99.99635% within ±1
Word count	6,625,231	99.99795% within ±5
Panel B: First independent claim validation		
First independent claim number	370,679	99.99811% exact
First independent claim length	370,679	99.48122% exact
First independent claim length	370,679	99.99002% within ±1
Panel C: Document-level validation		
Historical/reconstructed documents matched	379,680	—
Historical documents unmatched	120	—
Reconstruction-only documents	658	—
Independent claim count	379,680	99.07896% exact
Dependent claim count	379,680	98.76791% exact
Independent minimum word count	376,183	99.98671% exact
Independent average word count	376,183	99.97714% exact
Independent total word count	376,183	99.98565% exact
Dependent minimum word count	367,423	99.98993% exact
Dependent average word count	367,423	99.98748% exact
Dependent total word count	375,002	99.98880% exact

Notes: The table compares claims reconstructed from the 2014 USPTO Patent Application Publication Full-Text files with the historical USPTO claims data. Claim-level records are matched using publication number and claim number after imposing common restrictions on usable claim identifiers. Text-statistic comparisons exclude matched historical canceled claims for which the historical text measures are unavailable. The large set of reconstruction-only claims primarily reflects canceled claims; 84.24 percent of reconstruction-only claims are identified as canceled.

Panel B constructs the first non-canceled independent claim separately in each source. Panel C compares document-level statistics. Agreement for document-level word measures is reported conditional on the historical and reconstructed files containing the same number of the relevant independent or dependent claims. The historical claim-level and document-level files are not mechanically consistent with one another, so document-level claim-count disagreement need not imply a corresponding claim-level classification error.

Table 7: Validation of Reconstructed Patent-Grant Claims, 2014

Validation measure	Comparison N	Agreement / result
Panel A: Claim-level validation		
Reconstructed utility-patent claims matched	4,978,864	99.67674%
Reconstruction-only utility-patent claims	16,147	0.32326%
Independent/dependent classification	4,978,864	99.99998% exact
Word count	4,978,864	99.98891% exact
Word count	4,978,864	99.99827% within ±1
Word count	4,978,864	99.99950% within ±5
Panel B: First independent claim validation		
First independent claim number	300,101	99.99900% exact
First independent claim length	300,101	99.98967% exact
First independent claim length	300,101	99.99034% within ±1
Panel C: Document-level validation		
Reconstructed utility-patent documents matched	300,671	99.67776%
Reconstruction-only utility-patent documents	972	0.32224%
Independent claim count	300,671	99.87927% exact
Dependent claim count	300,671	99.87927% exact
Total claim count	300,671	100.00000% exact
Independent minimum word count	300,308	99.98701% exact
Independent average word count	300,308	99.98535% exact
Independent total word count	300,308	99.98535% exact
Dependent minimum word count	295,744	99.99865% exact
Dependent average word count	295,744	99.99391% exact
Dependent total word count	300,308	99.99401% exact

Notes: The table compares claims reconstructed from the 2014 USPTO Patent Grant Full-Text files with the historical USPTO grant claims data. Claim-level records are matched using application number, patent number, and claim number. The common patent-number matching key is numeric, so the validation exercise is restricted to utility patents; this is a restriction on the validation comparison rather than on the reconstructed grant data.

The historical grant files do not contain a usable grant-year variable, so reconstructed 2014 claims and patents are matched against the full historical grant universe. Historical-only records therefore include grants from other years and cannot be interpreted as unmatched 2014 observations. Reported coverage is consequently measured relative to the reconstructed 2014 utility-patent comparison universe. Panel B constructs the first independent claim separately in each source. Document-level word-count agreement in Panel C is conditional on the two sources containing the same number of relevant independent or dependent claims.

$$E_a^{\text{adj}} = 1000\left(\bar{L} + \bar{\varepsilon}_a\right). \tag{9}$$

The resulting measure removes common grant-cohort differences while preserving cross-art-unit differences in historical litigation exposure. The preferred analysis sample retains four-digit examiner art units with at least 500 eligible historical grants. No requirement is imposed that an art unit be observed in every year of the historical period.

Table 8 summarizes the historical sample and the resulting exposure distribution. The full four-digit historical universe contains 1,573,621 eligible grants, of which 10,587 were litigated during 2003–2011, corresponding to a raw litigation rate of approximately 6.73 patents per 1,000 grants. Of the 719 four-digit examiner art units observed in the historical universe, 427 satisfy the 500-grant requirement. These preferred art units contain 1,543,382 historical grants, or approximately 98.1 percent of the historical four-digit grant population.

Table 8: Historical Litigation Exposure: Sample and Distribution

	Value
Panel A: Historical exposure sample	
Eligible four-digit grants, 2003–2011	1,573,621
Historically litigated grants	10,587
Raw litigation rate per 1,000 grants	6.728
Four-digit examiner art units	719
Preferred examiner art units ($\geq$ 500 grants)	427
Historical grants in preferred art units	1,543,382
Share of historical grants retained (%)	98.08
Panel B: Grant-year-adjusted exposure	
Number of preferred art units	427
Mean	8.085
Standard deviation	6.695
25th percentile	3.860
Median	6.242
75th percentile	10.392
Minimum	-2.423
Maximum	39.497
Correlation with raw exposure	0.952

Notes: Historical exposure is constructed from regular utility patents granted between 2003 and 2011 and district-court litigation observed during the same period. Panel B reports the unweighted distribution across the 427 preferred four-digit examiner art units. Grant-year-adjusted exposure is an index expressed in per-1,000-grant units rather than a literal litigation rate. The preferred sample requires at least 500 eligible historical grants but does not require an art unit to appear in every historical year.

The adjusted exposure distribution is right-skewed, as shown in Figure 8. The median preferred

art unit has an adjusted exposure value of approximately 6.24 per 1,000 grants, compared with a mean of approximately 8.09. A one-standard-deviation increase in the standardized exposure measure used in the regressions corresponds to 6.695 units of the grant-year-adjusted exposure index per 1,000 historical grants. Because the adjusted measure is constructed from residualized litigation propensity, a small number of art units have slightly negative adjusted values. These values should therefore be interpreted as values of a grant-year-adjusted exposure index rather than literal negative litigation rates.

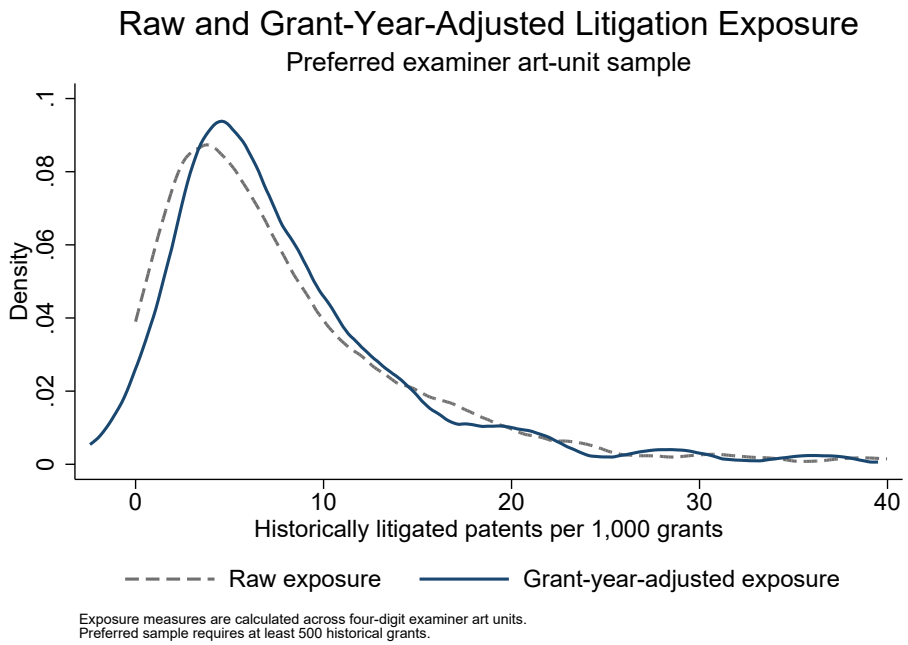


Figure 8: Distribution of Historical Litigation Exposure

Notes: The figure shows the distribution of grant-year-adjusted historical litigation exposure across the 427 preferred four-digit examiner art units. Exposure is expressed in adjusted litigated patents per 1,000 historical grants. The preferred sample contains examiner art units with at least 500 eligible grants during 2003–2011.

Figure 9 compares the raw litigation rate with the grant-year-adjusted measure for individual examiner art units. The two measures are closely related, with a correlation of approximately 0.95. The grant-year adjustment therefore does not substantially reorder the cross-sectional exposure distribution. Instead, it primarily removes common differences in observed litigation propensity associated with the grant cohort.

Finally, the baseline empirical specifications standardize E_a^{adj} across the preferred examiner-art-unit sample,

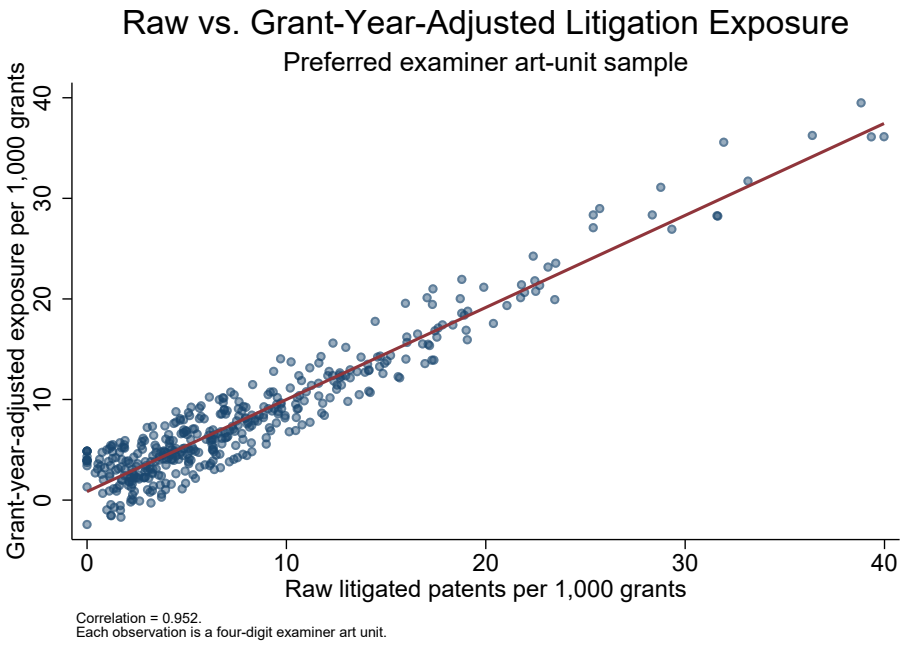


Figure 9: Raw and Grant-Year-Adjusted Historical Litigation Exposure

Notes: Each observation represents one preferred four-digit examiner art unit. The horizontal axis reports raw historical litigation exposure, measured as litigated patents per 1,000 grants. The vertical axis reports the grant-year-adjusted exposure index used to construct the standardized historical exposure measure. The correlation between the two measures is 0.952.

$$Exposure_a = \frac{E_a^{\text{adj}} - \overline{E^{\text{adj}}}}{s(E^{\text{adj}})}. \tag{10}$$

Accordingly, the coefficient on the interaction between historical exposure and the post-AIA indicator in the main specifications is scaled to a one-standard-deviation difference in historical exposure. In the preferred sample, one such standard deviation corresponds to approximately 6.695 units of the adjusted exposure index per 1,000 historical grants.

B.1 Additional Descriptive Exposure Gradients

Figure 10 presents the stage-specific versions of the raw exposure-gradient figure reported in the main text, including pointwise 95 percent confidence intervals. Each coefficient is estimated separately within filing cohort and therefore reports the contemporaneous cross-sectional relationship between historical exposure and the indicated outcome rather than a coefficient normalized to an omitted filing cohort. As in the main-text descriptive figure, the publication, grant, and publication-to-grant-change panels refer to different underlying populations.

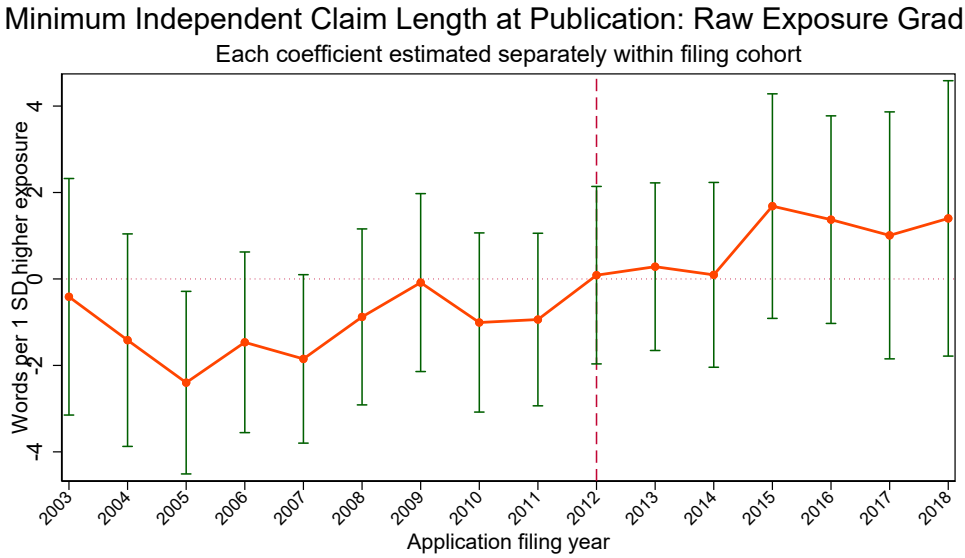
C Additional Robustness and Matched-Sample Evidence

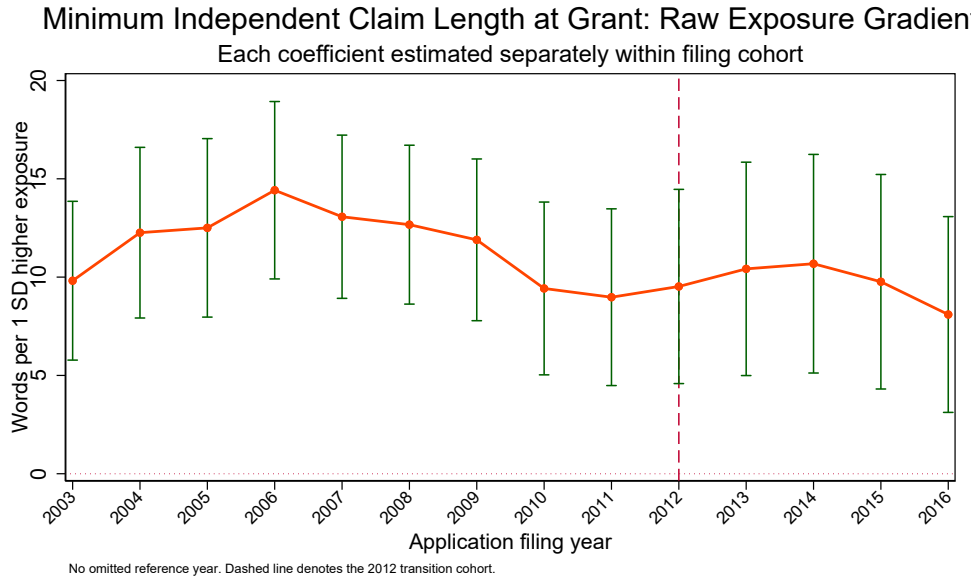
This appendix reports additional evidence supporting the robustness exercises discussed in Section 8. I first present the pooled stability table under different year requirements. I then report nonlinear specifications that replace continuous historical litigation exposure with fixed exposure groups. Finally, I present event studies estimated on matched publication–grant samples so that publication outcomes, prosecution-stage changes, and grant outcomes are measured for the same applications.

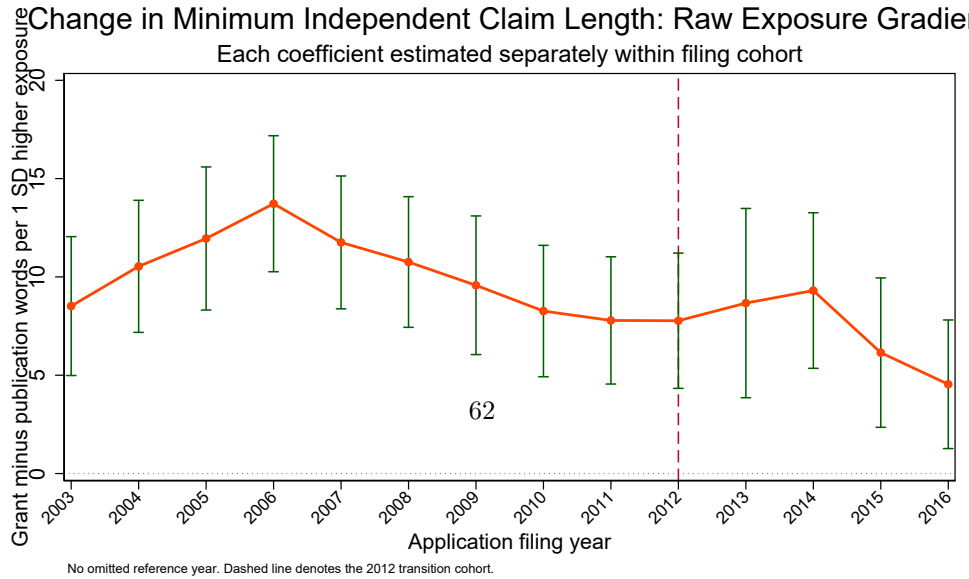
C.1 Historical Examiner-Art-Unit Stability

The preferred analysis requires an examiner art unit to contain at least 500 eligible historical grants during 2003–2011 but does not require the art unit to appear in every individual year of the historical period. Section 8.1 shows that the publication event-study path is similar as progressively stronger historical-presence requirements are imposed. Table 9 reports the corresponding pooled estimates and documents the sample attrition generated by these restrictions.

The estimated Exposure × Post coefficient remains between approximately 2.25 and 2.70 words


Minimum Independent Claim Length at Publication: Raw Exposure Gradie
Each coefficient estimated separately within filing cohort
Words per 1 SD higher exposure
-4
-2
0
2
4
2003
2004
2005
2006
2007
2008
2009
2010
2011
2012
2013
2014
2015
2016
2017
2018
Application filing year
No omitted reference year. Dashed line denotes the 2012 transition cohort.


Minimum Independent Claim Length at Grant: Raw Exposure Gradient
Each coefficient estimated separately within filing cohort
Words per 1 SD higher exposure
0
5
10
15
20
2003
2004
2005
2006
2007
2008
2009
2010
2011
2012
2013
2014
2015
2016
Application filing year
No omitted reference year. Dashed line denotes the 2012 transition cohort.


Change in Minimum Independent Claim Length: Raw Exposure Gradier
Each coefficient estimated separately within filing cohort
Grant minus publication words per 1 SD higher exposure
0
5
10
15
20
2003
2004
2005
2006
2007
2008
2009
2010
2011
2012
2013
2014
2015
2016
Application filing year
No omitted reference year. Dashed line denotes the 2012 transition cohort.

across all five specifications. Requiring presence in at least six of the nine historical years retains 372 examiner art units and about 91 percent of the preferred application sample. The eight-of-nine requirement retains 297 art units and about 79 percent of applications, while requiring presence in every historical year leaves 240 art units and approximately 69 percent of the preferred sample. The stability of the coefficient despite this substantial sample attrition supports using the less restrictive 500-grant criterion as the baseline.

Table 9: Robustness to Historical Examiner-Art-Unit Presence

Historical-presence requirement	Estimate	SE	Art units	Applications	Retained (%)
No annual-presence restriction	2.255***	0.731	427	4,164,106	100.00
At least 6 of 9 years	2.444***	0.752	372	3,807,363	91.43
At least 7 of 9 years	2.363***	0.778	335	3,589,420	86.20
At least 8 of 9 years	2.507***	0.807	297	3,274,085	78.63
All 9 years	2.699***	0.873	240	2,876,767	69.08

Notes: The table reports the coefficient on standardized historical litigation exposure interacted with the post-AIA indicator for minimum independent claim length at publication. All specifications require at least 500 eligible historical grants during 2003–2011 and differ only in the minimum number of historical years in which the examiner art unit must be observed. Historical litigation exposure is standardized using the preferred 500+ unbalanced examiner-art-unit sample and the same exposure scale is retained in every specification. Regressions include examiner-art-unit and application-filing-year fixed effects, omit the 2012 transition cohort, weight art-unit–filing-year cells by the number of underlying applications, and cluster standard errors by examiner art unit. The final column reports the percentage of the preferred publication minimum-ICL sample retained. Standard errors are reported in the third column. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

C.2 Nonlinear Historical Litigation Exposure

The preferred specifications model historical litigation exposure continuously. Table 10 examines whether this linear specification masks substantial differences across the exposure distribution. The table reports three alternative representations of historical exposure. Panel A divides examiner art units into quintiles. Panel B isolates the upper decile by separately identifying approximately the 90th–95th percentile and the top 5 percent. Panel C compares the upper 10 percent of examiner art units with the remaining 90 percent. All groups are fixed using the pre-AIA cross-sectional distribution of examiner-art-unit exposure.

The publication minimum-ICL results show the clearest increasing pattern across quintiles. Relative to the lowest exposure quintile, the estimated post-AIA difference is 0.36 words in the second quintile, 2.35 words in the third, 3.50 words in the fourth, and 4.74 words in the highest quintile. The fourth- and fifth-quintile estimates are statistically distinguishable from zero. The corresponding FICL estimates are less monotonic, although the largest positive estimate again occurs in the

highest quintile.

The upper-tail comparisons are substantially less precise. The estimated post-AIA difference between the top 5 percent and the bottom 90 percent is 7.97 words for minimum ICL and 9.37 words for FICL, but both estimates have large standard errors. Likewise, the direct top-10-versus-bottom-90 comparisons are positive but imprecise. These specifications therefore provide little evidence that the publication result can be reduced to a discrete break among only the most exposed examiner art units.

The grant and prosecution-change columns reinforce the more cautious interpretation of those stages. Their nonlinear coefficients are less stable across exposure groups, and several of the largest differences appear in intermediate rather than highest exposure categories. This is consistent with the event-study evidence that the grant and prosecution outcomes contain substantial pre-existing dynamics and should not be interpreted as independent causal estimates of the AIA response.

C.3 Matched Publication–Grant Event Studies

The stage-specific results in the main text use the largest available population for each outcome. The publication sample therefore includes applications regardless of whether they later issue, whereas the grant sample contains issued patents. To verify that differences across stages are not driven mechanically by these changing populations, I construct matched samples containing only applications for which the relevant claim-length measure is observed at both publication and grant.

For minimum independent claim length, the matched sample contains 2,401,596 applications distributed across 5,462 examiner-art-unit by filing-year cells. The corresponding FICL sample contains 2,397,378 applications and 5,461 cells. Both samples contain the same 427 examiner art units used in the preferred analysis. Within each matched sample, publication claim length, grant claim length, and their difference are constructed for the same applications and collapsed using identical cell weights.

Figures 11–13 report the matched-sample minimum-ICL event studies. The publication series exhibits relatively small and irregular exposure differences before the AIA. The coefficients become substantially larger in the later post-AIA cohorts, reaching approximately 2.80 words in both 2015 and 2016 relative to the 2011 exposure gradient.

The prosecution-change series displays a different pattern. Historical exposure is associated with substantially larger publication-to-grant increases in minimum ICL during several early pre-AIA

Table 10: Nonlinear Historical Litigation Exposure Specifications

	Publication		Grant		Publication-to-Grant Change	
	Minimum ICL	First ICL	Minimum ICL	First ICL	Minimum ICL	First ICL
Panel A: Exposure quintiles						
Q2 × Post	0.360	-1.481	0.240	-0.579	-1.719	-1.123
	(1.633)	(2.010)	(2.365)	(2.545)	(2.134)	(2.184)
Q3 × Post	2.346	-2.671	-0.331	-2.105	-3.177	-0.354
	(2.472)	(4.300)	(2.681)	(3.110)	(2.374)	(3.125)
Q4 × Post	3.505**	2.329	-4.781**	-5.887**	-6.927***	-7.011***
	(1.473)	(1.634)	(2.333)	(2.419)	(2.120)	(2.219)
Q5 × Post	4.739**	3.780*	0.678	0.008	-4.169*	-3.688
	(1.854)	(2.203)	(2.646)	(2.799)	(2.337)	(2.452)
Panel B: Upper-tail exposure groups						
90th–95th percentile × Post	1.702	1.980	0.864	0.653	-1.176	-1.192
	(3.962)	(5.187)	(4.056)	(4.610)	(3.978)	(3.957)
Top 5% × Post	7.967*	9.371*	8.669	10.508	0.242	0.258
	(4.101)	(5.056)	(7.642)	(8.296)	(5.916)	(6.398)
Panel C: Upper-decile comparison						
Top 10% × Post	4.412	5.179	3.642	4.160	-0.680	-0.684
	(2.954)	(3.785)	(3.656)	(4.038)	(3.295)	(3.408)
Underlying observations	4,164,106	4,156,739	2,596,100	2,596,089	2,401,596	2,397,378
Examiner-art-unit FE	Yes	Yes	Yes	Yes	Yes	Yes
Filing-year FE	Yes	Yes	Yes	Yes	Yes	Yes
Examiner-art-unit clusters	427	427	427	427	427	427

Notes: Each panel reports coefficients from a separate set of regressions. Panel A divides examiner art units into five fixed historical-exposure quintiles, with the lowest quintile omitted. Panel B uses the bottom 90 percent as the omitted group and separately identifies examiner art units in approximately the 90th–95th percentile and the upper 5 percent of the historical-exposure distribution. Panel C compares approximately the upper 10 percent with the bottom 90 percent. Publication regressions use filing cohorts through 2018; grant and matched-change regressions use filing cohorts through 2016. The 2012 transition cohort is omitted. All specifications include examiner-art-unit and filing-year fixed effects, weight cells by the number of underlying applications or patents, and cluster standard errors by examiner art unit. Standard errors are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

cohorts. This differential declines toward 2011 and eventually becomes negative in 2015 and 2016. The grant-stage coefficients combine these two components, producing the positive pre-AIA grant gradients documented in the main results. Because publication ICL, prosecution change, and grant ICL are measured for exactly the same applications, these differences across panels reflect changes in where the grant-stage exposure relationship emerges rather than differences in sample composition.

The corresponding FICL event studies are reported in Figures 14–16. They exhibit the same broad division across stages. The matched publication series contains relatively little systematic movement in the pre-AIA exposure relationship and becomes more positive in the later post-AIA cohorts. The 2015 and 2016 publication coefficients are approximately 2.90 and 2.95 words, respectively. In contrast, the prosecution-change series contains large positive exposure differences in several early cohorts before declining over time and becoming negative by 2015–2016. The grant-stage FICL series inherits these pre-existing prosecution dynamics.

The similarity between the matched and unrestricted publication results also indicates that the main publication finding is not generated by changes in which published applications subsequently receive patents. The matched exercises are nevertheless best interpreted as stage and mechanism evidence. Restricting attention to applications that eventually issue conditions on a later prosecution outcome and therefore does not provide a preferable replacement for the unrestricted publication specification.

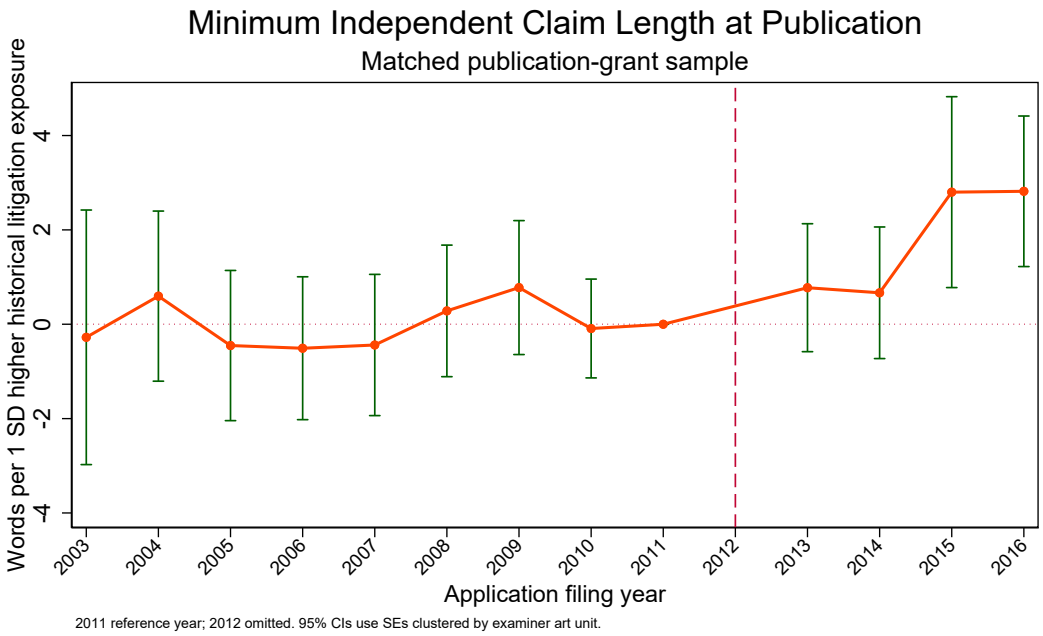


Figure 11: Matched-Sample Event Study: Minimum Independent Claim Length at Publication

Notes: The sample contains applications with minimum independent claim length observed at both publication and grant. Historical litigation exposure is standardized at the examiner-art-unit level. The 2011 filing cohort is the reference year and 2012 is omitted as the transition cohort. The regression includes examiner-art-unit and filing-year fixed effects, weights cells by the number of underlying matched applications, and reports 95 percent confidence intervals using standard errors clustered by examiner art unit.

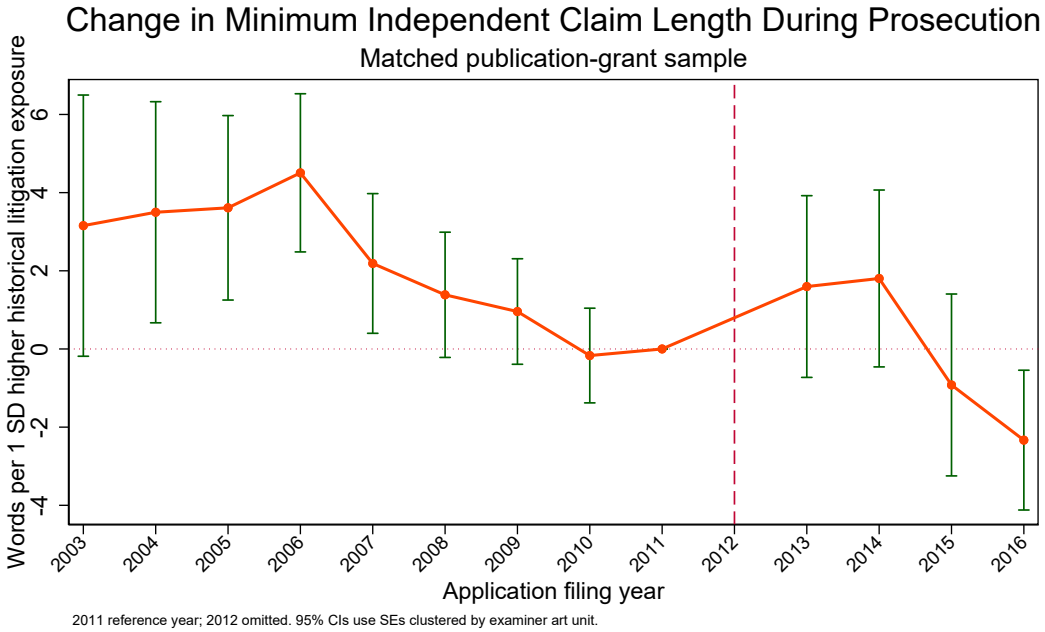


Figure 12: Matched-Sample Event Study: Change in Minimum Independent Claim Length During Prosecution

Notes: The outcome is grant-stage minimum independent claim length minus publication-stage minimum independent claim length for the same matched applications used in Figure 11. The 2011 filing cohort is the reference year and 2012 is omitted. The regression includes examiner-art-unit and filing-year fixed effects, uses matched-sample cell weights, and reports 95 percent confidence intervals using standard errors clustered by examiner art unit.

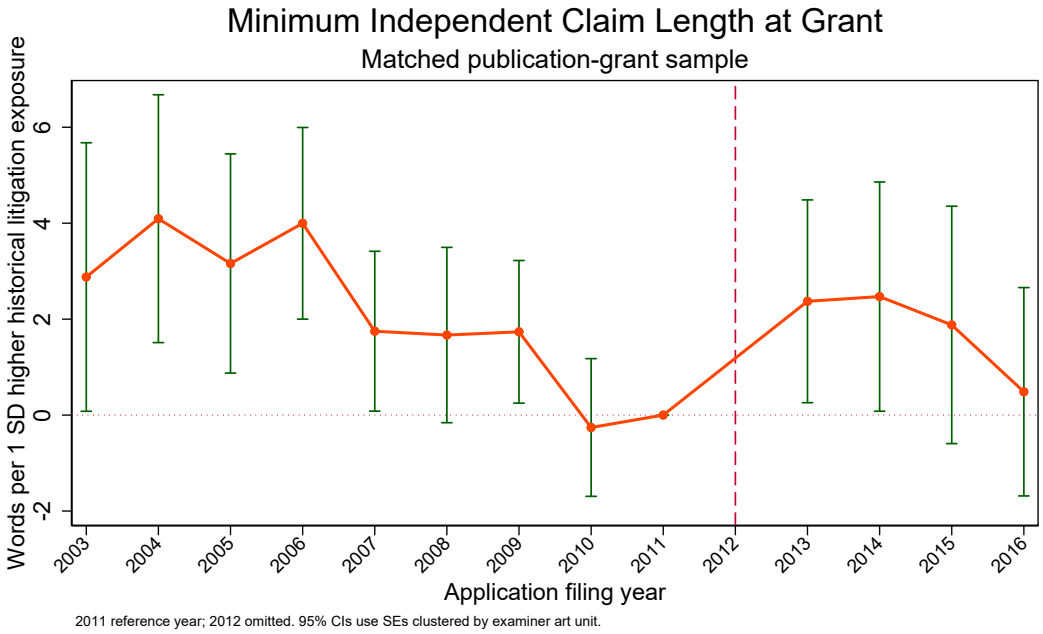


Figure 13: Matched-Sample Event Study: Minimum Independent Claim Length at Grant

Notes: The sample and weights are identical to those used for the matched publication and prosecution-change event studies. By construction, the grant-stage event-study coefficient in each filing cohort equals the publication-stage coefficient plus the prosecution-change coefficient, up to numerical precision. The 2011 filing cohort is the reference year and 2012 is omitted. Confidence intervals use standard errors clustered by examiner art unit.

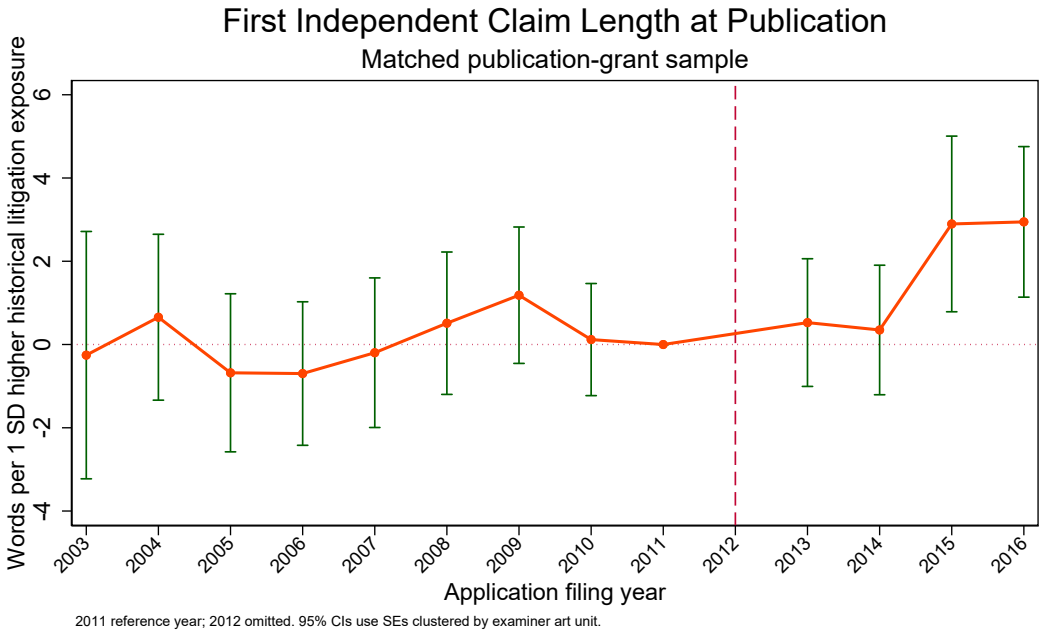


Figure 14: Matched-Sample Event Study: First Independent Claim Length at Publication

Notes: The sample contains applications with first independent claim length observed at both publication and grant. The 2011 filing cohort is the reference year and 2012 is omitted as the transition cohort. The regression includes examiner-art-unit and filing-year fixed effects, weights cells by the number of underlying matched applications, and reports 95 percent confidence intervals using standard errors clustered by examiner art unit.

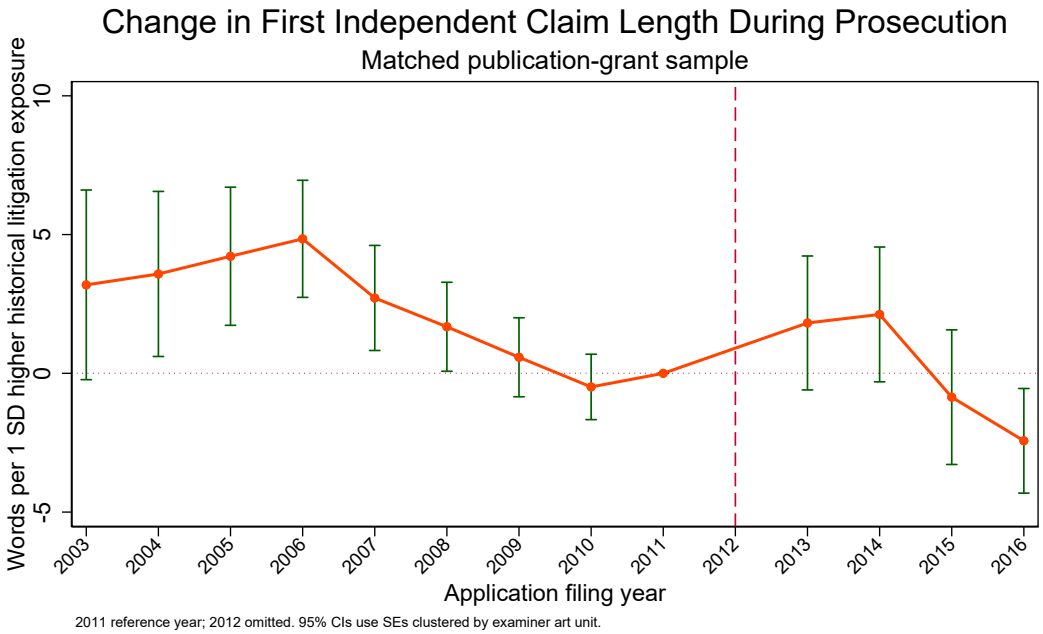


Figure 15: Matched-Sample Event Study: Change in First Independent Claim Length During Prosecution

Notes: The outcome is grant-stage first independent claim length minus publication-stage first independent claim length for the same matched applications used in Figure 14. The 2011 filing cohort is the reference year and 2012 is omitted. The regression includes examiner-art-unit and filing-year fixed effects, uses matched-sample cell weights, and reports 95 percent confidence intervals using standard errors clustered by examiner art unit.

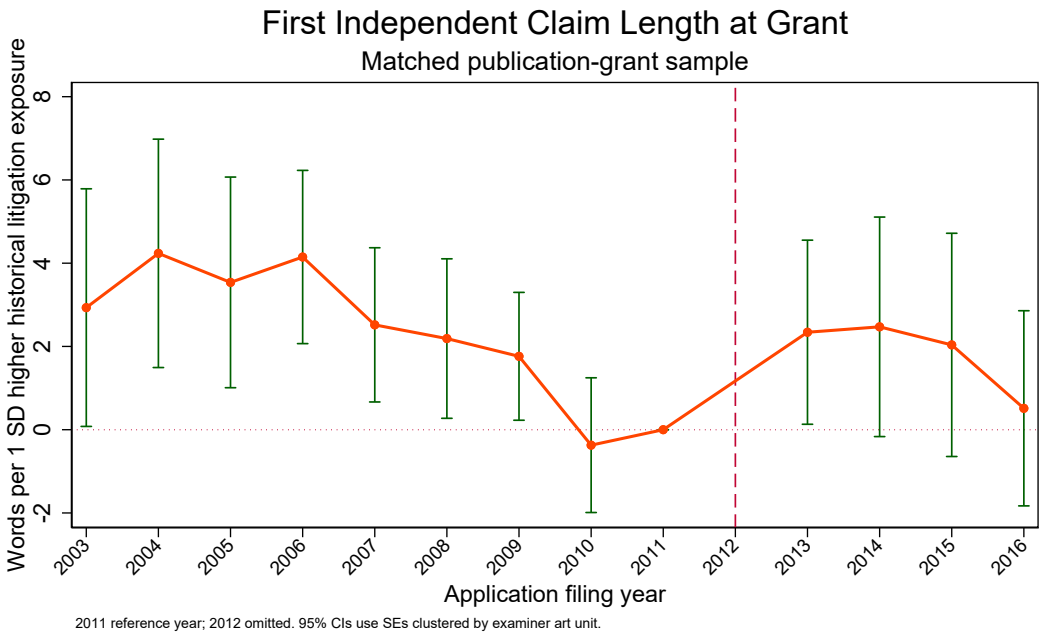


Figure 16: Matched-Sample Event Study: First Independent Claim Length at Grant

Notes: The sample and weights are identical to those used for the matched publication and prosecution-change FICL event studies. By construction, the grant-stage event-study coefficient in each filing cohort equals the publication-stage coefficient plus the prosecution-change coefficient, up to numerical precision. The 2011 filing cohort is the reference year and 2012 is omitted. Confidence intervals use standard errors clustered by examiner art unit.